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(54) **BODY WEIGHT SUPPORT**

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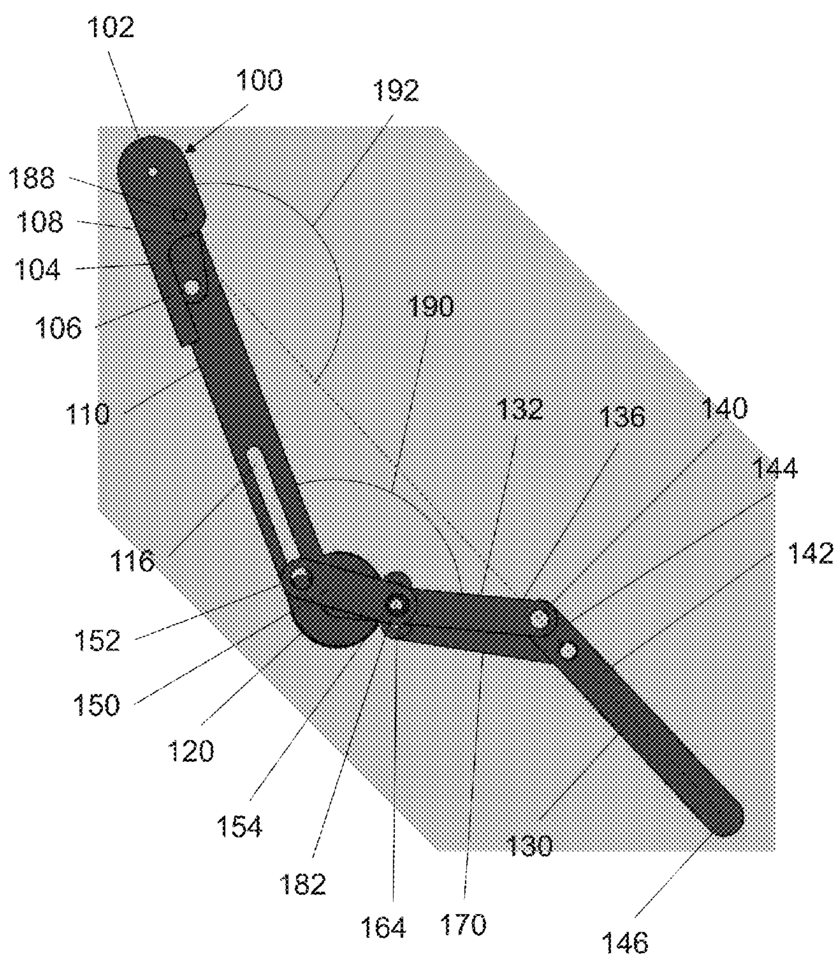
(57) **ABSTRACT**

The present disclosure relates to a body weight support, comprising: a first member configured for location against a first portion of a limb of a user, the first portion of the limb being on a first side of a joint of the user; a second member configured for location against a second portion of the limb on a second side of the joint of the user, wherein the second member is rotatable relative to the first member about a pivot, the second member comprising: a first portion proximate to the pivot; and a second portion extending from the first portion, distal to the pivot; wherein the second portion is movable from a first configuration in which a first angle between the first portion and the first member is less than a second angle between the second portion and the first member, to a second configuration in which the difference between the first angle and the second angle is less than the difference between the first angle and the second angle in the first configuration; and a load bearing element configured to provide a resistance to rotation of the second member relative to the first member.



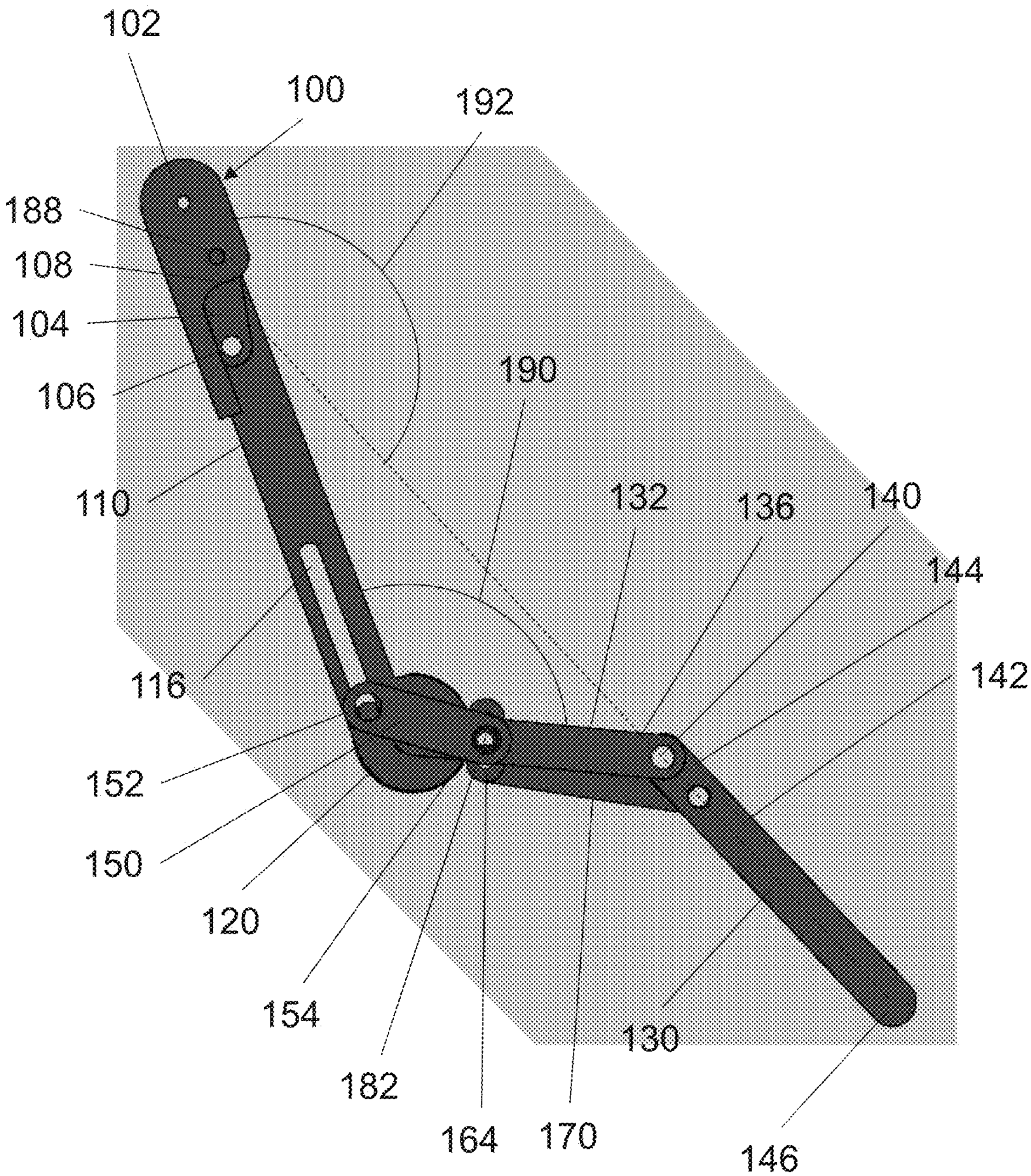

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FIG. 1

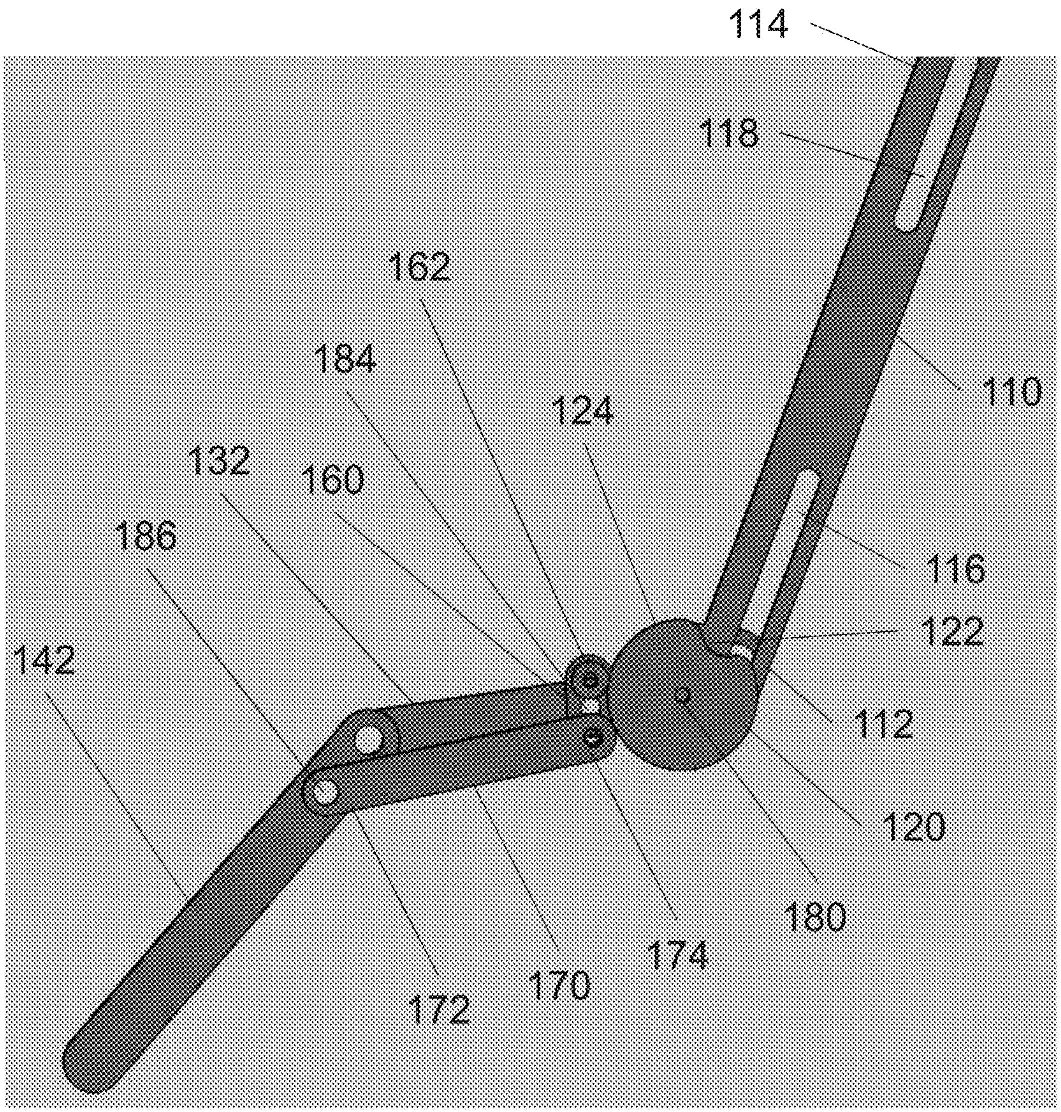
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FIG. 2

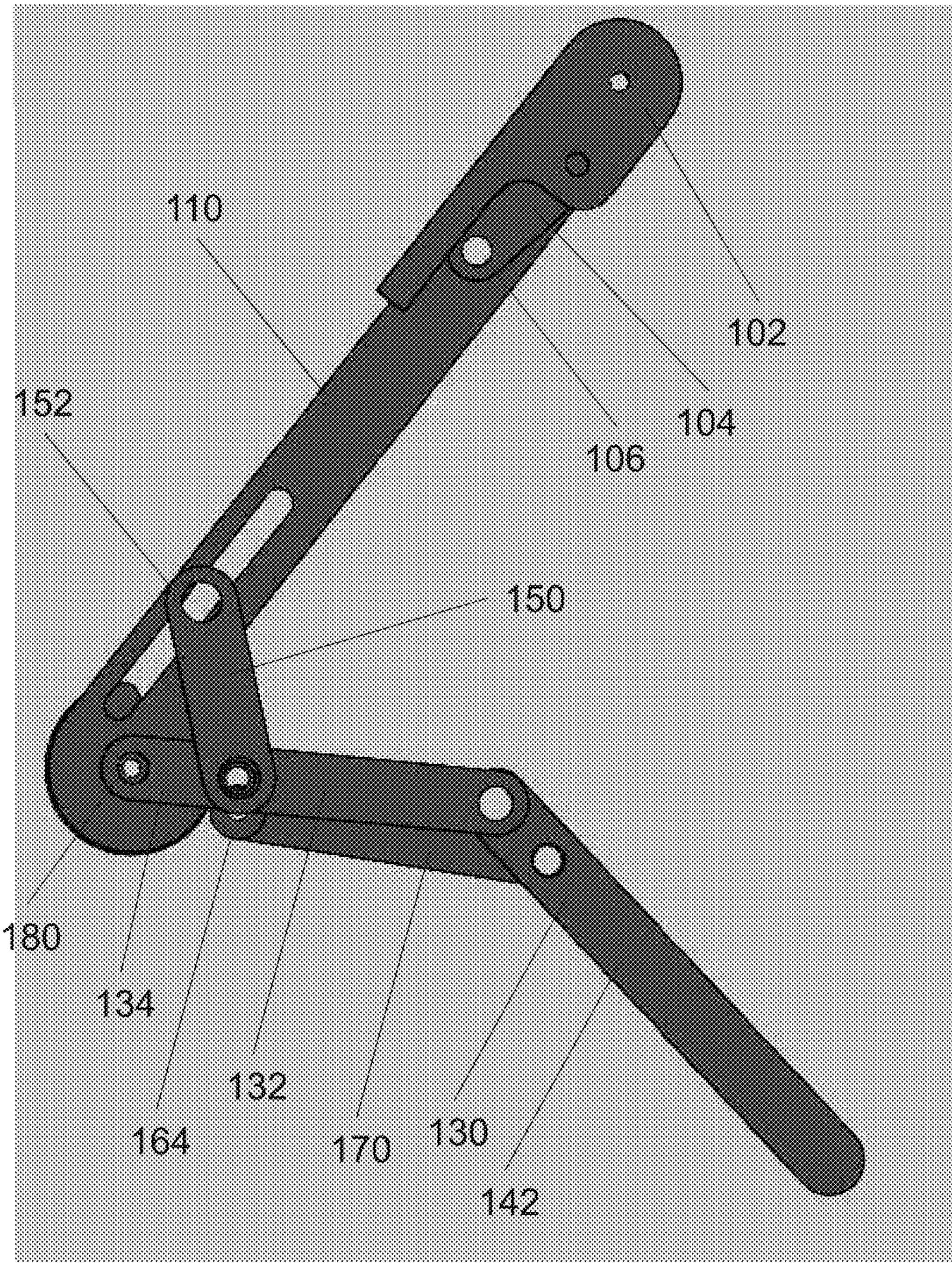

FIG. 3A

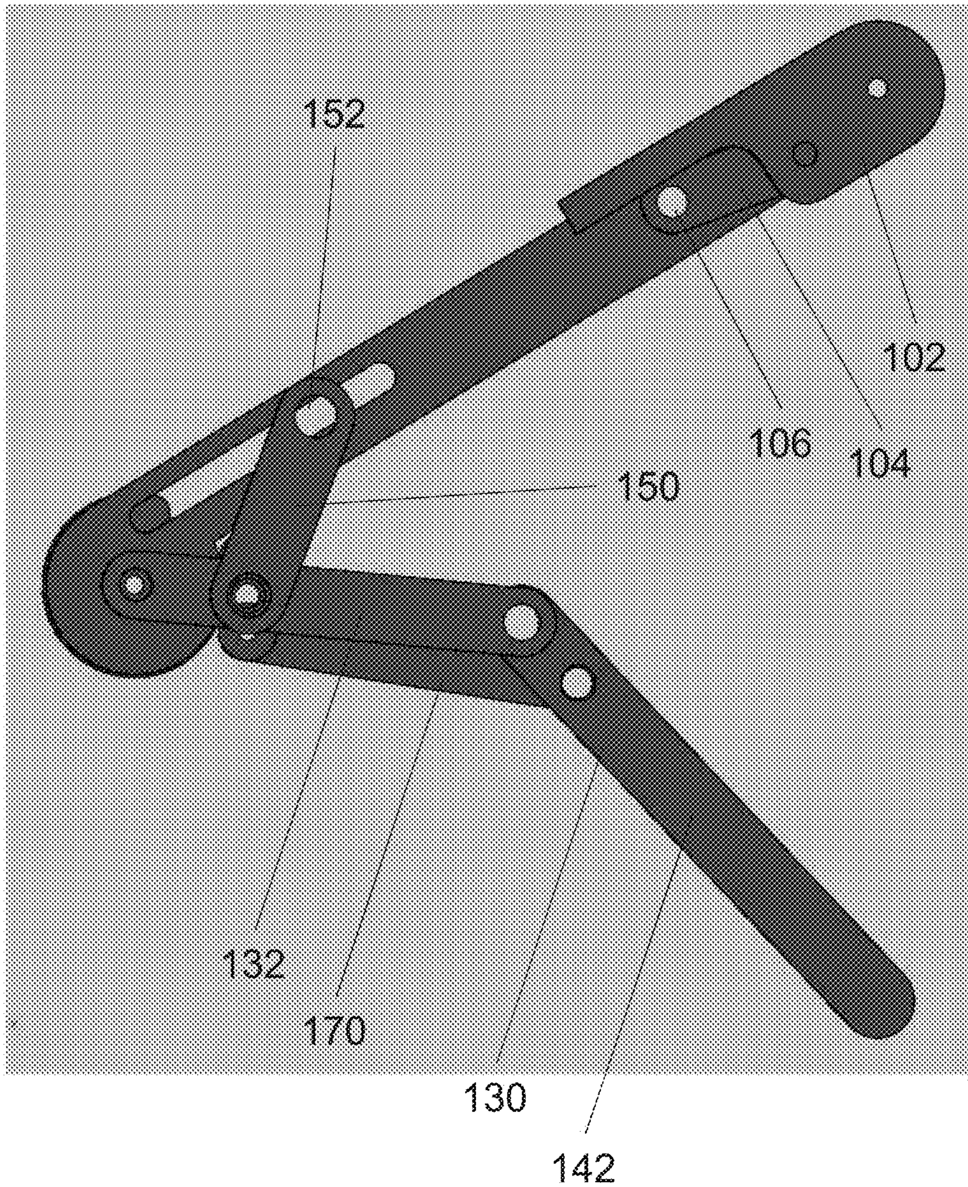

FIG. 3B

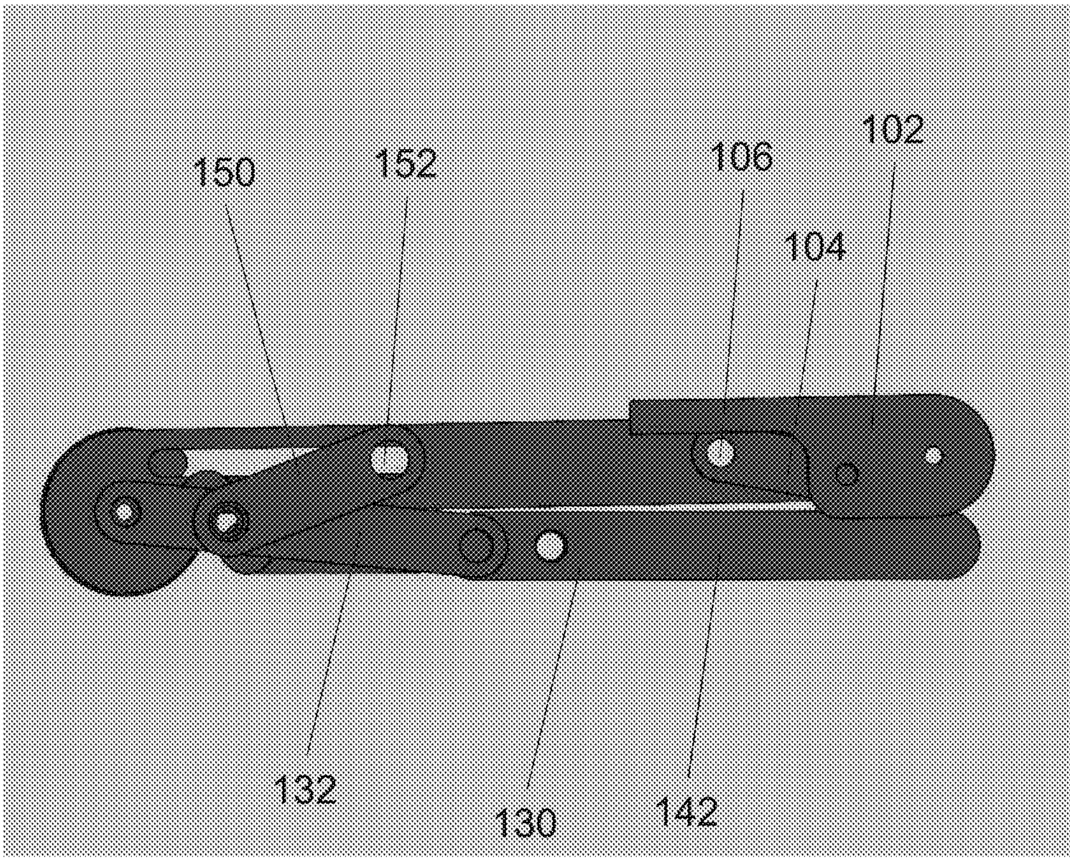

FIG. 3C

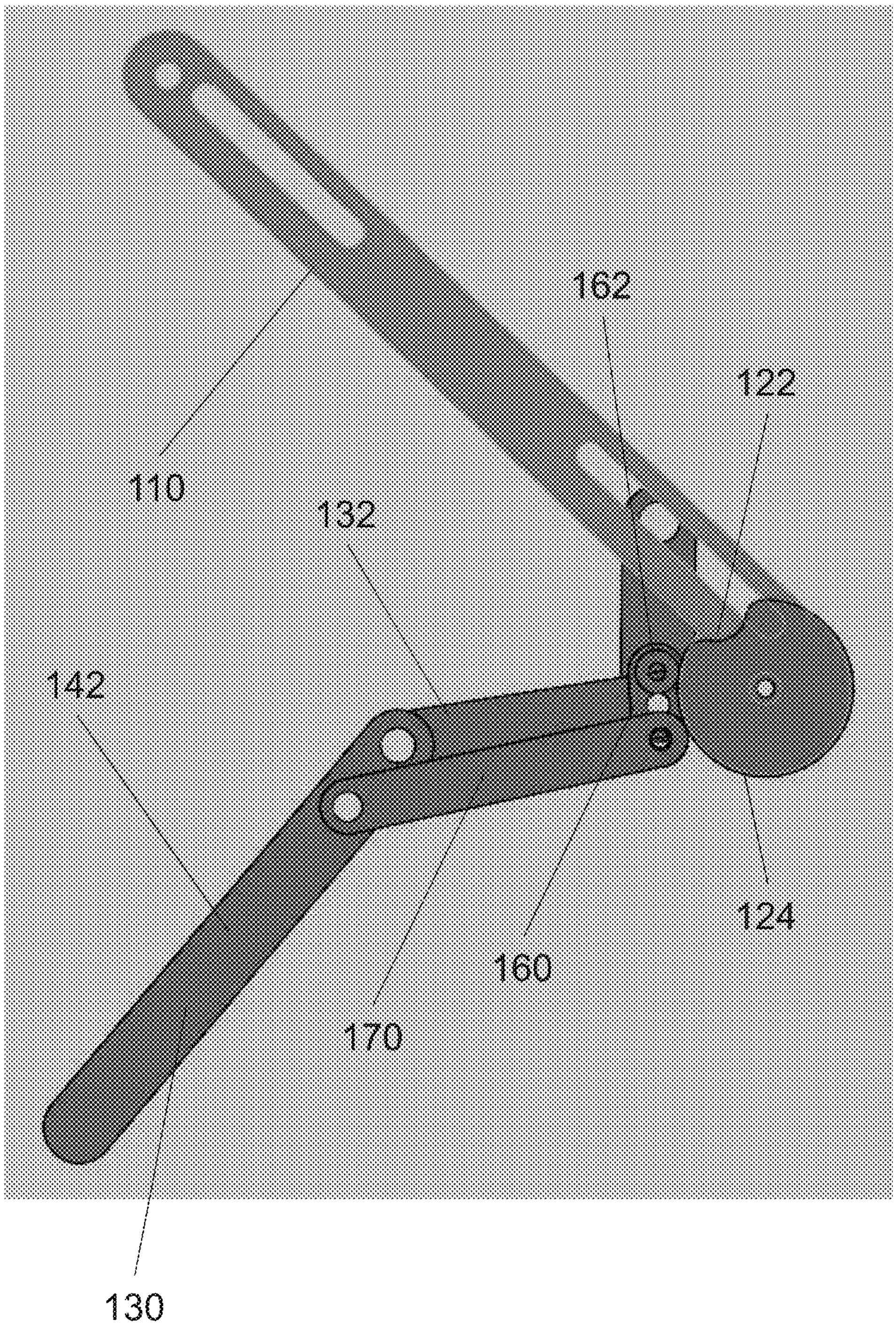

FIG. 4A

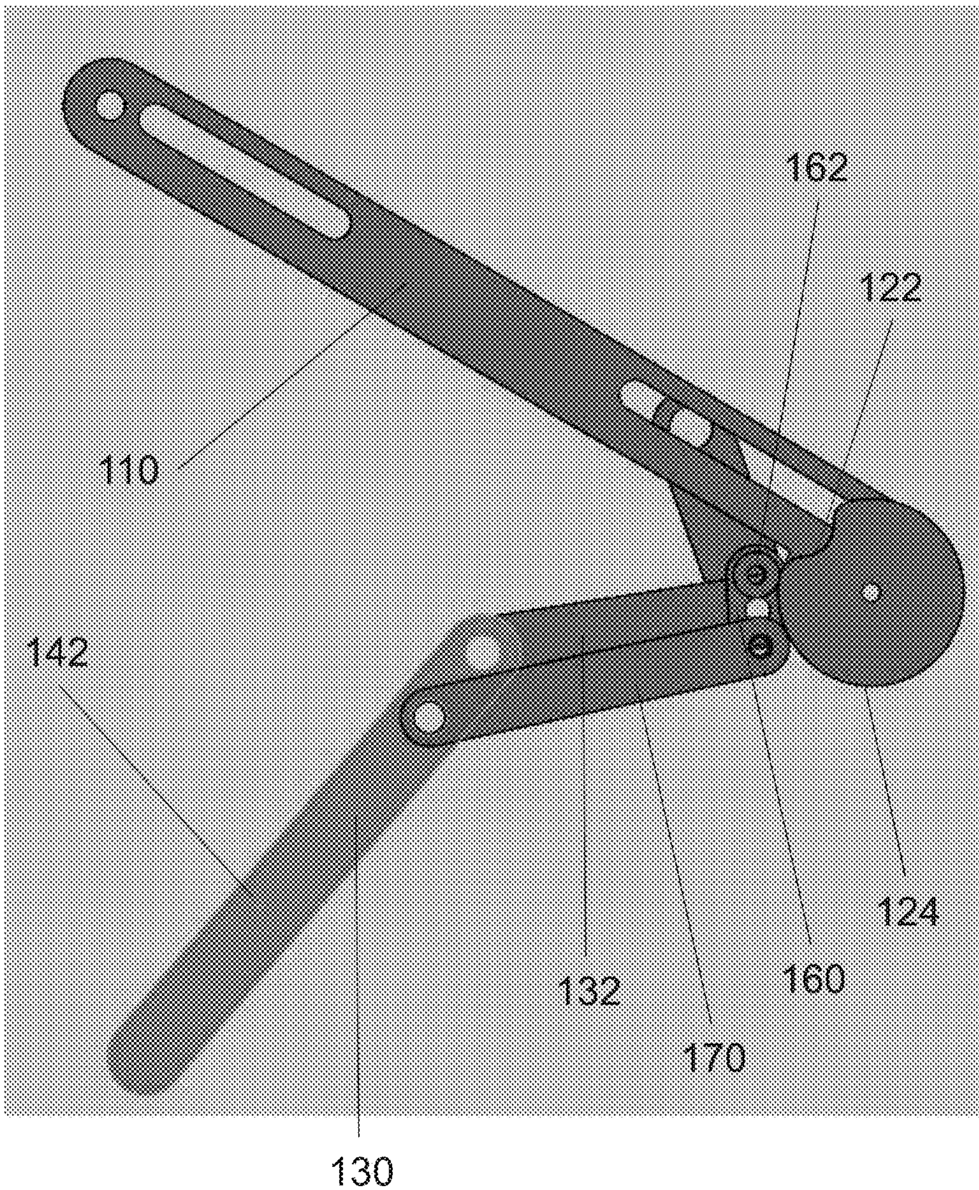

FIG. 4B

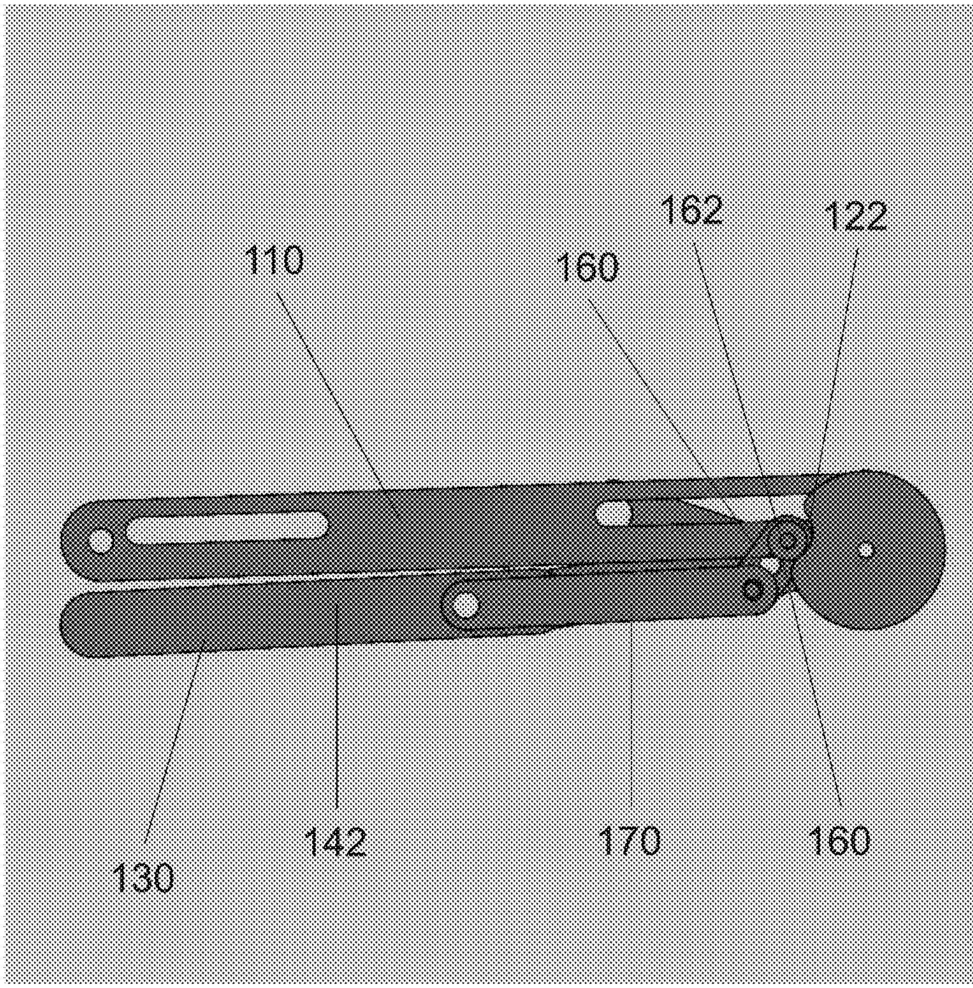

FIG. 4C

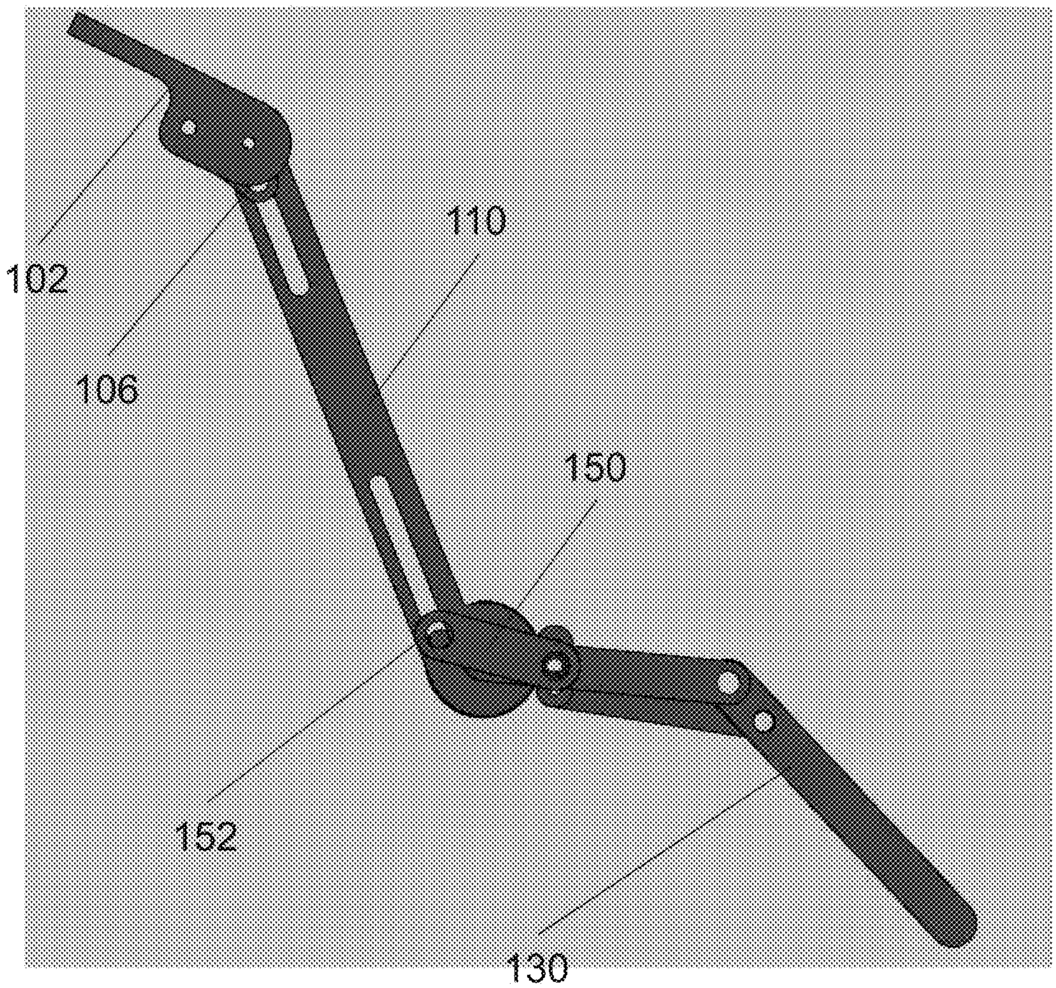

FIG. 5A

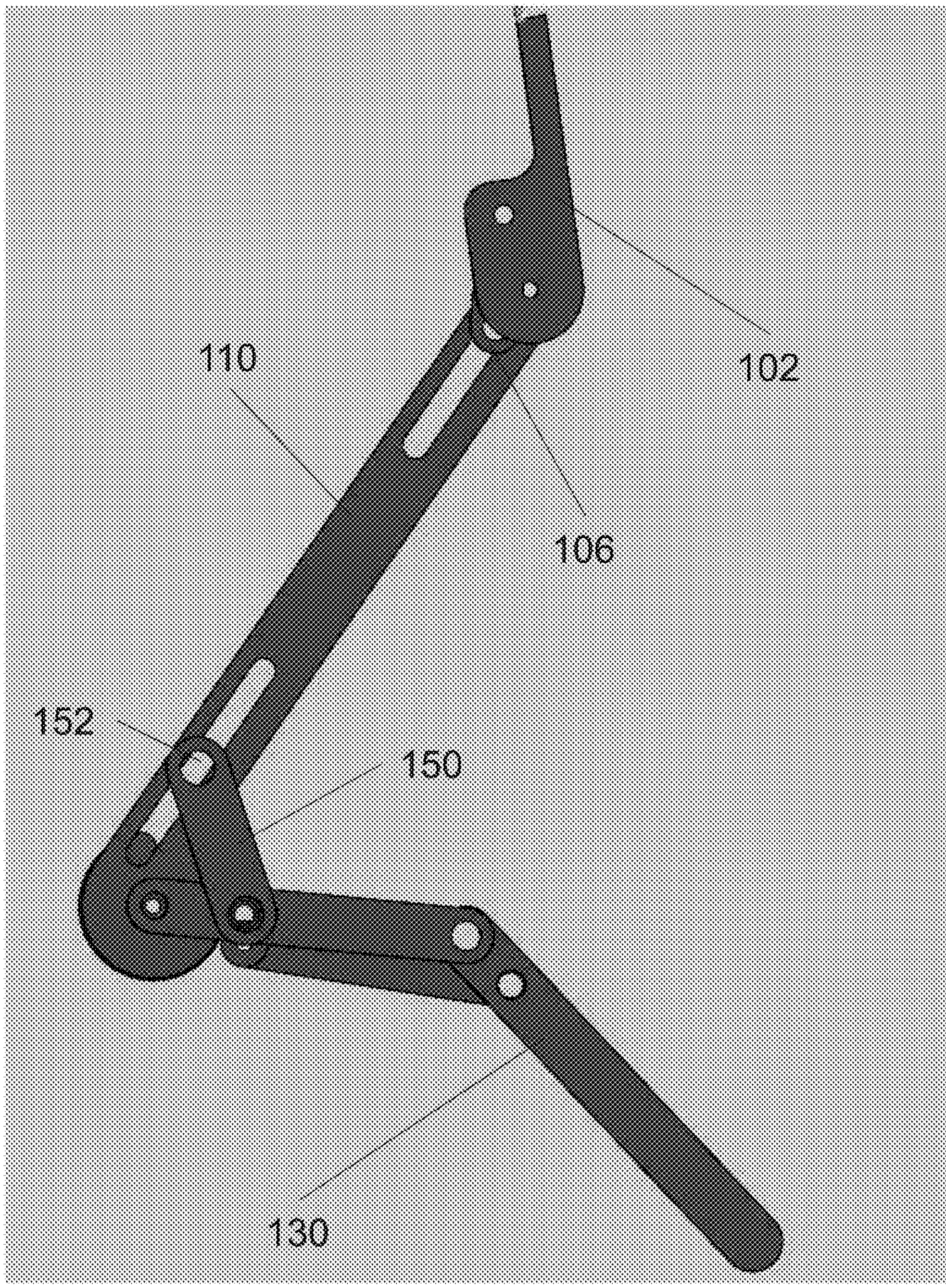

FIG. 5B

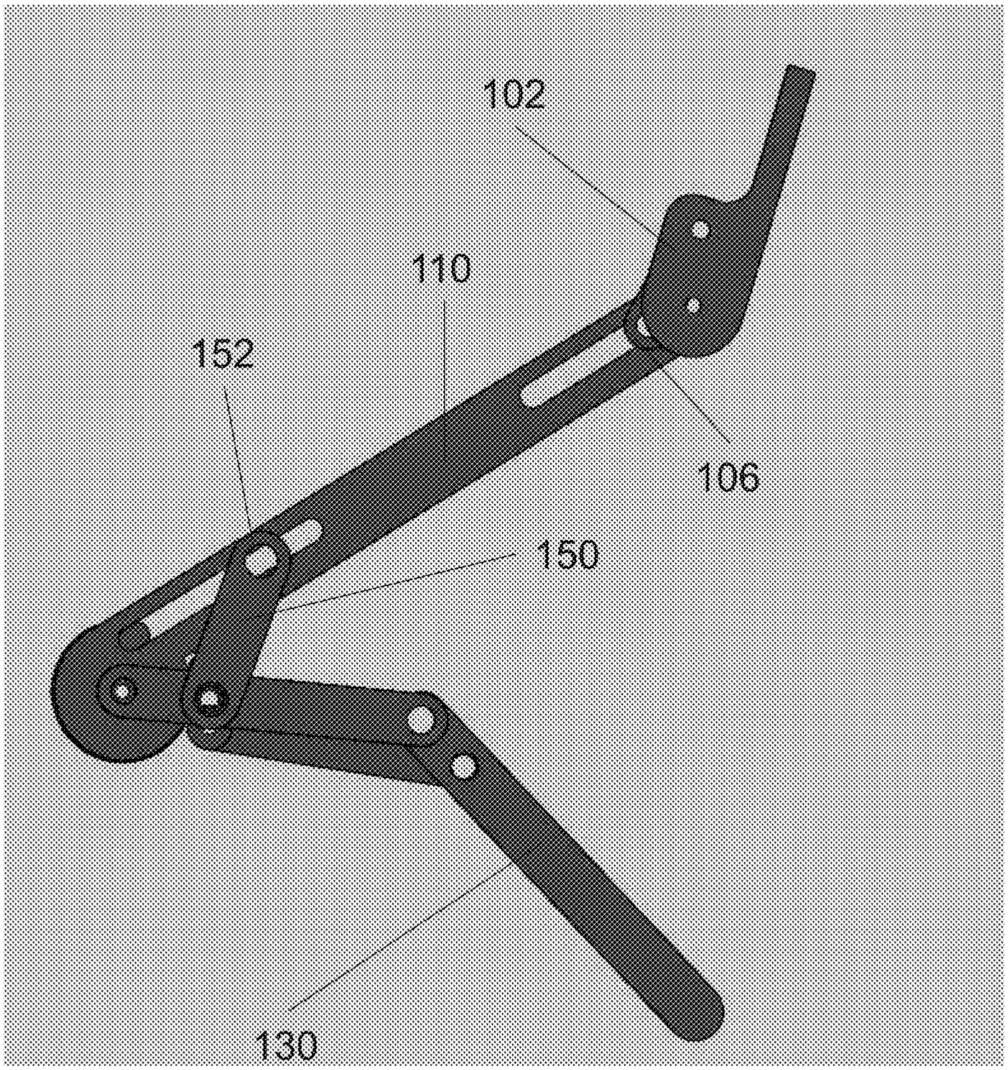

FIG. 5C

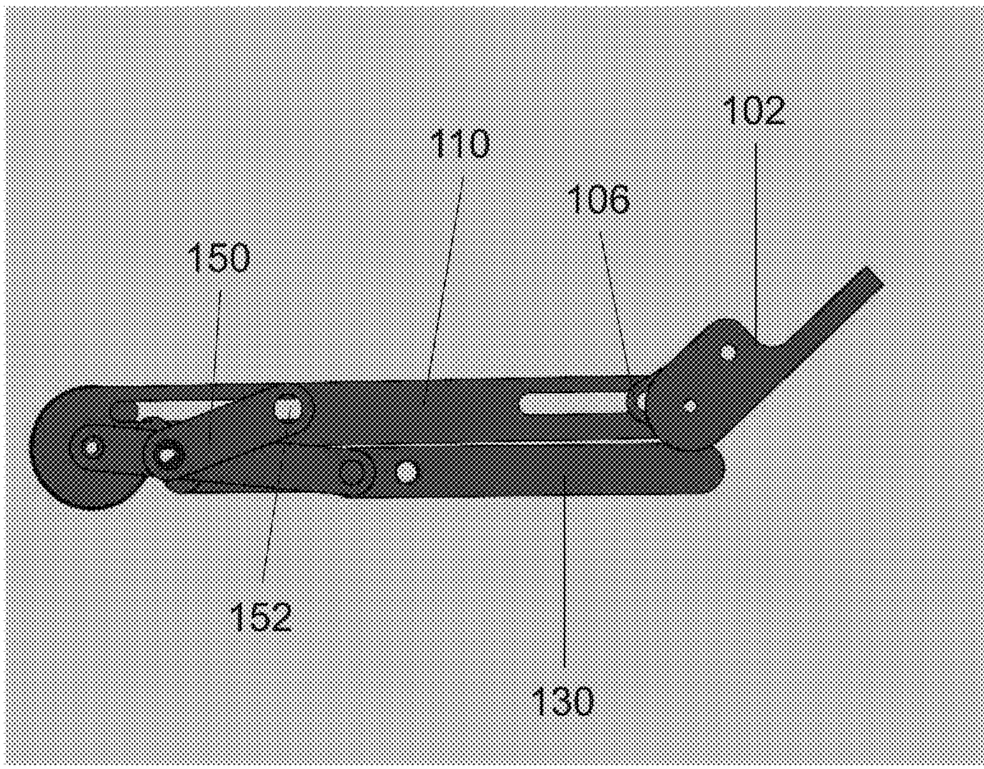

FIG. 5D

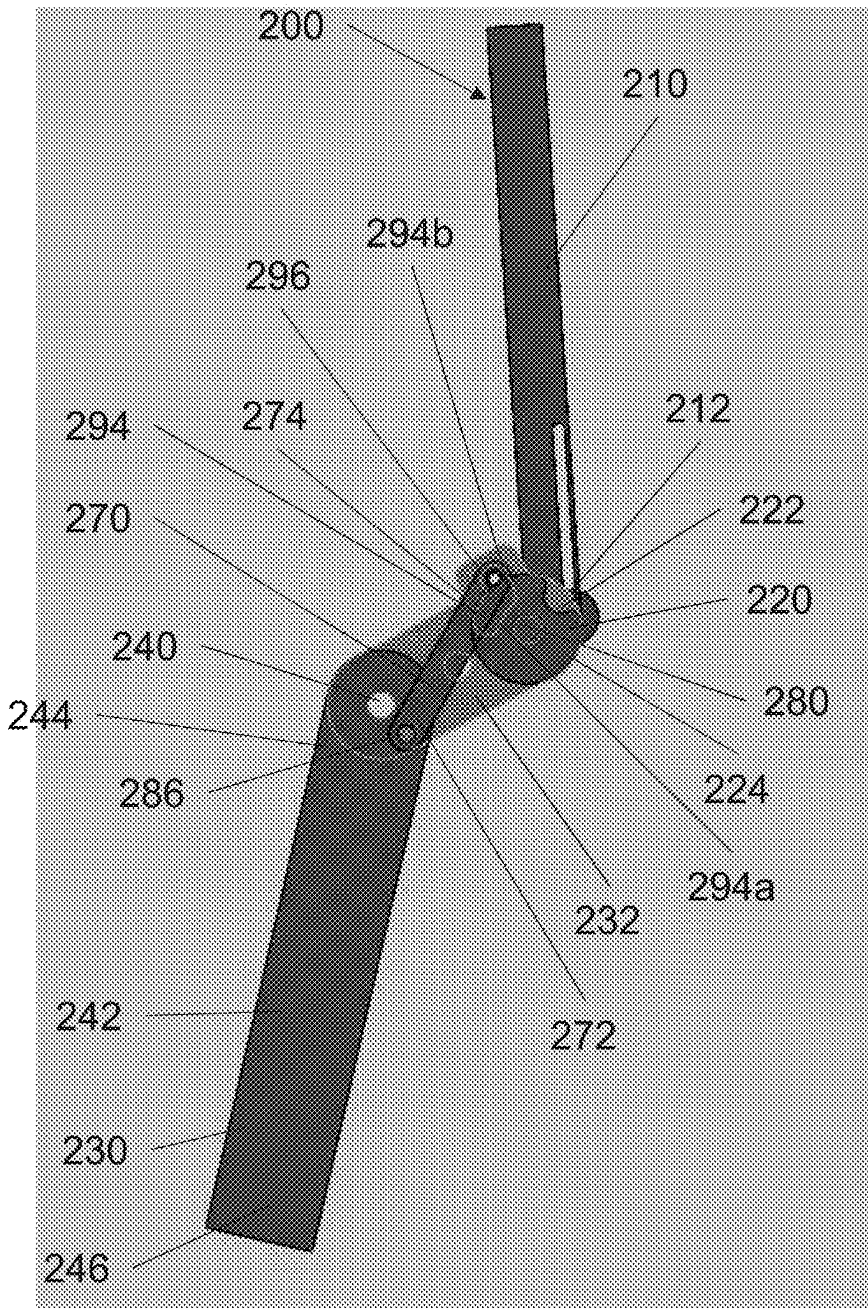
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FIG. 6A

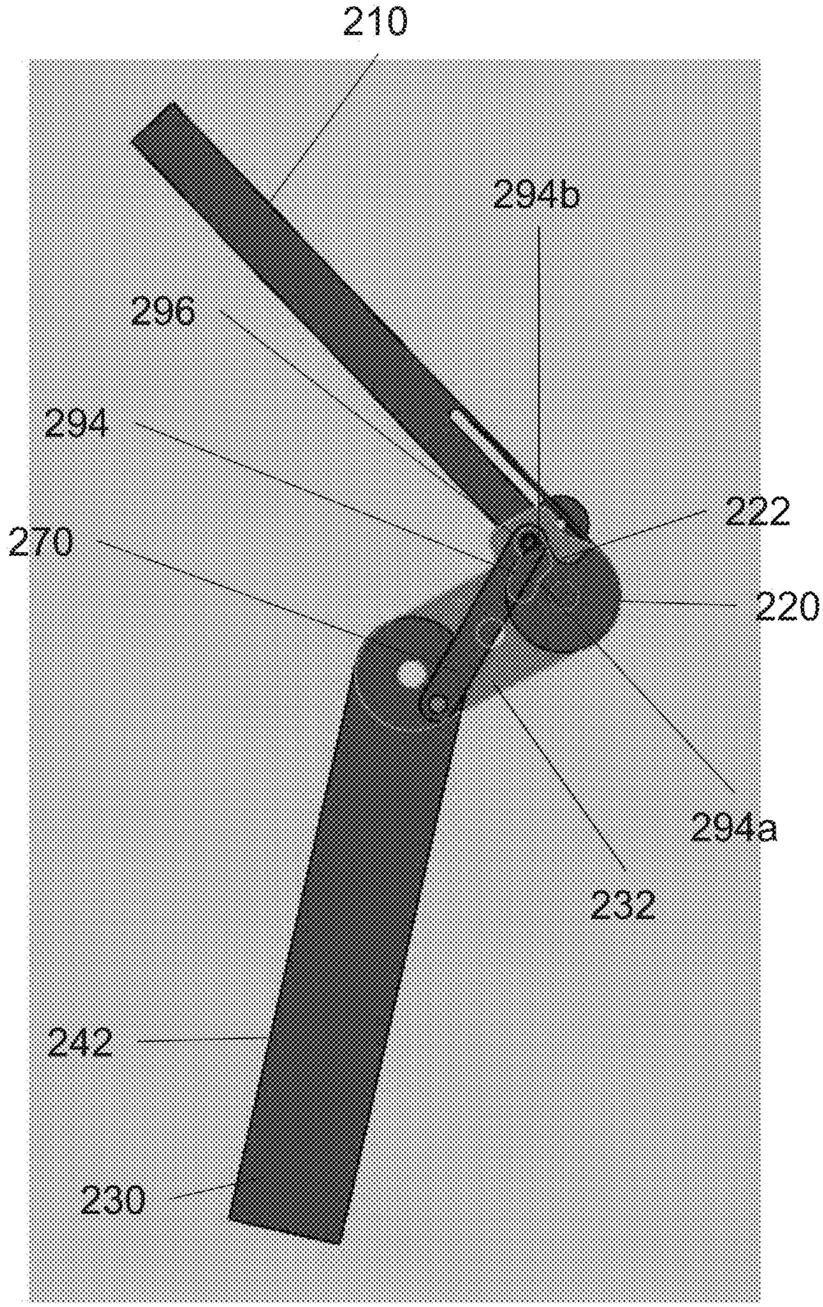

FIG. 6B

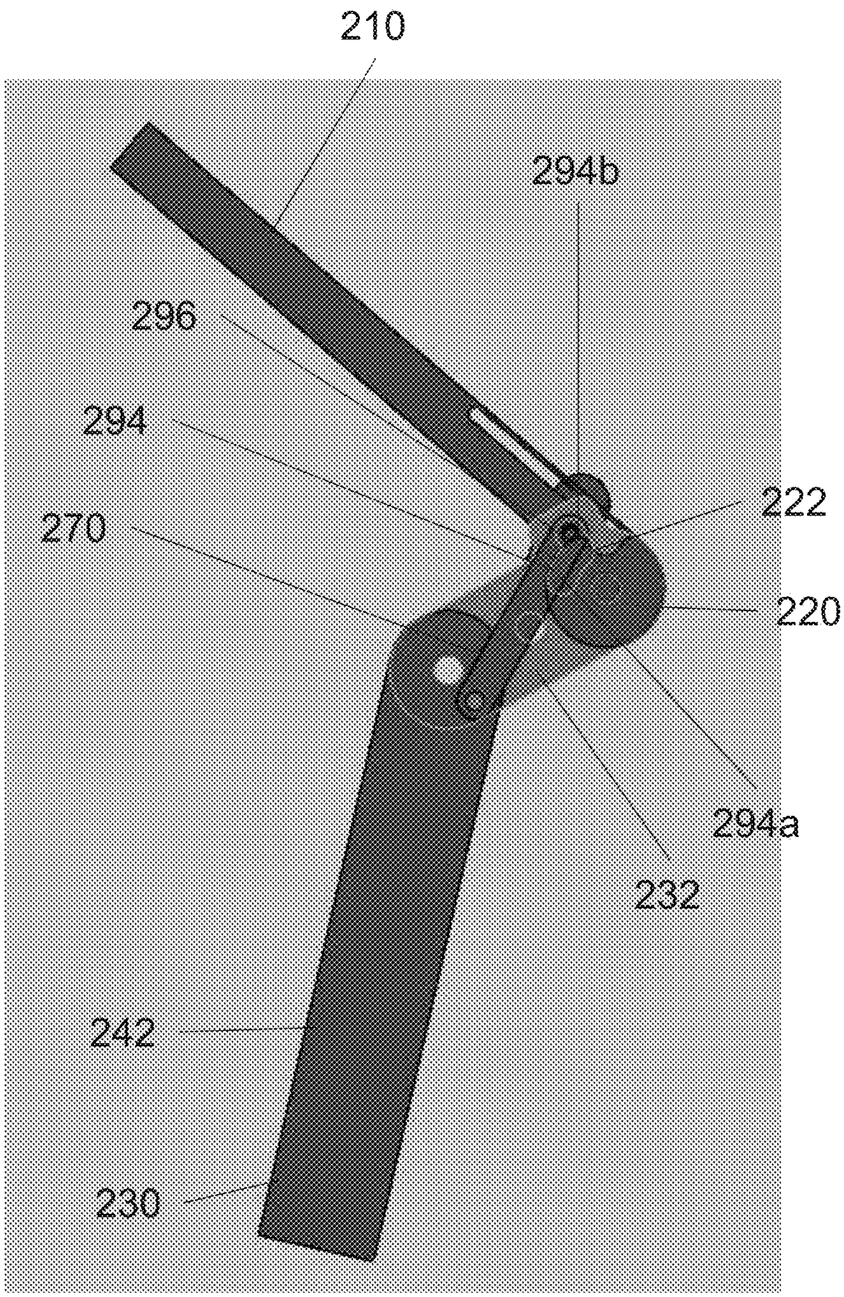

FIG. 6C

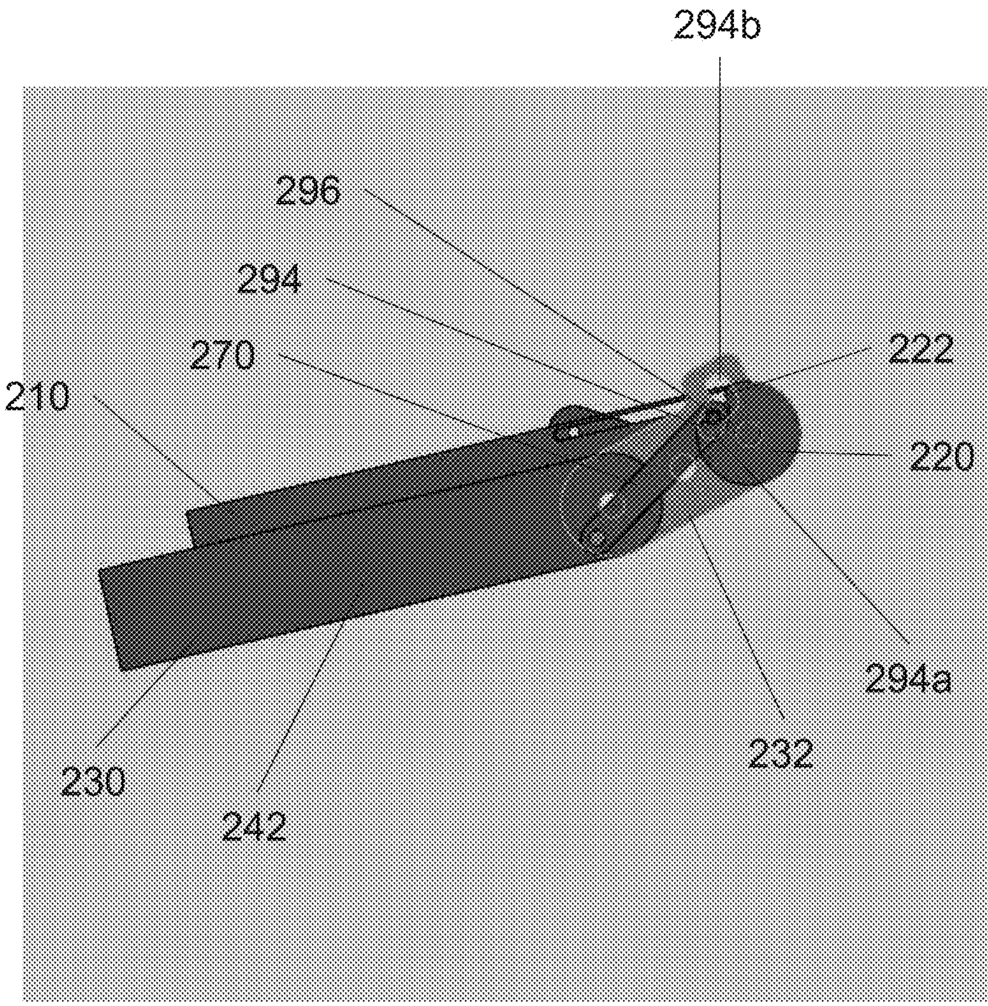

FIG. 6D

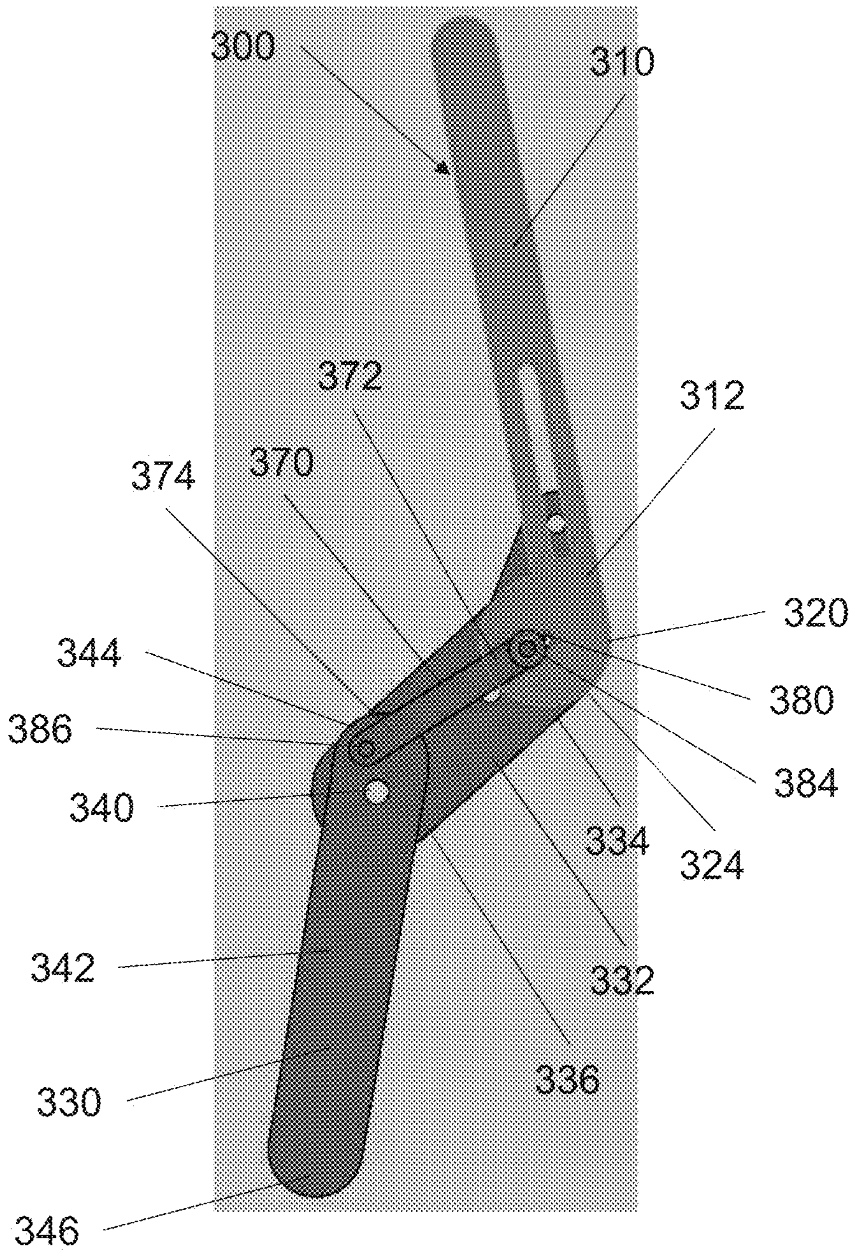
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FIG. 7A

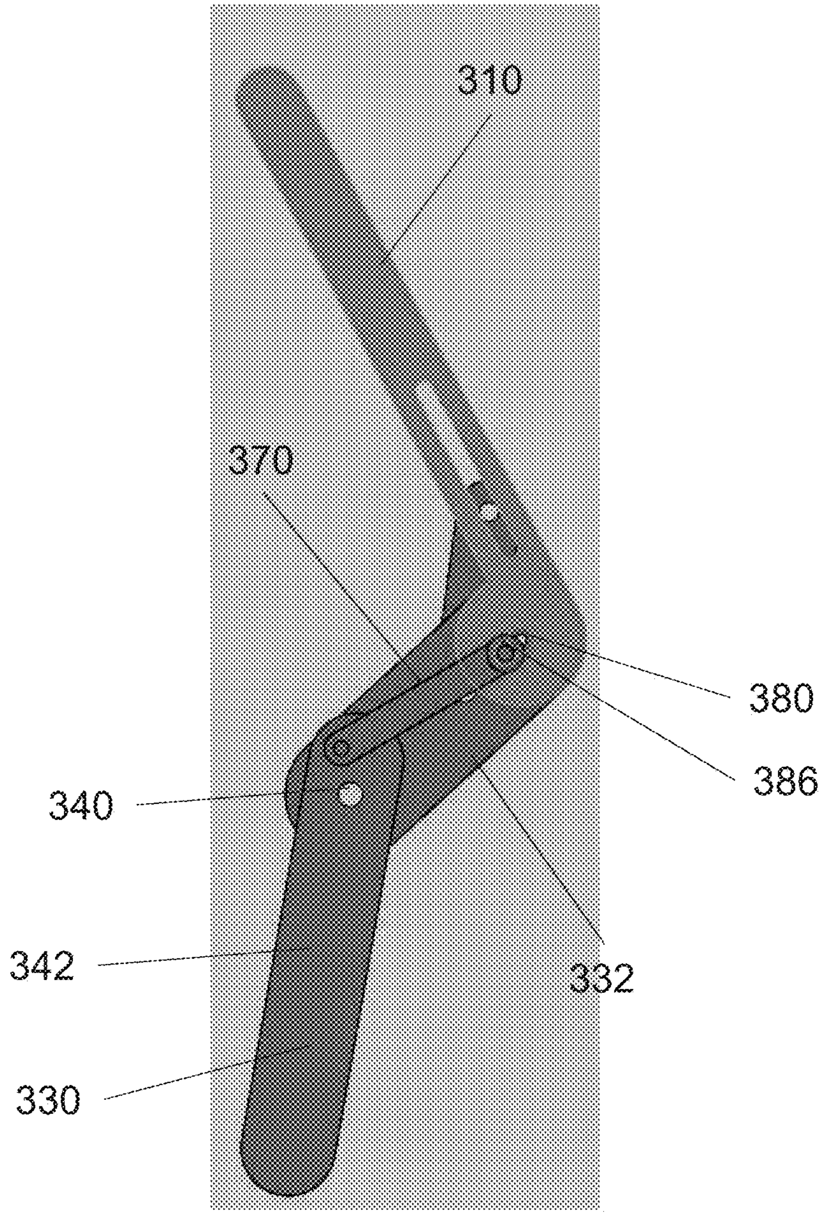

FIG. 7B

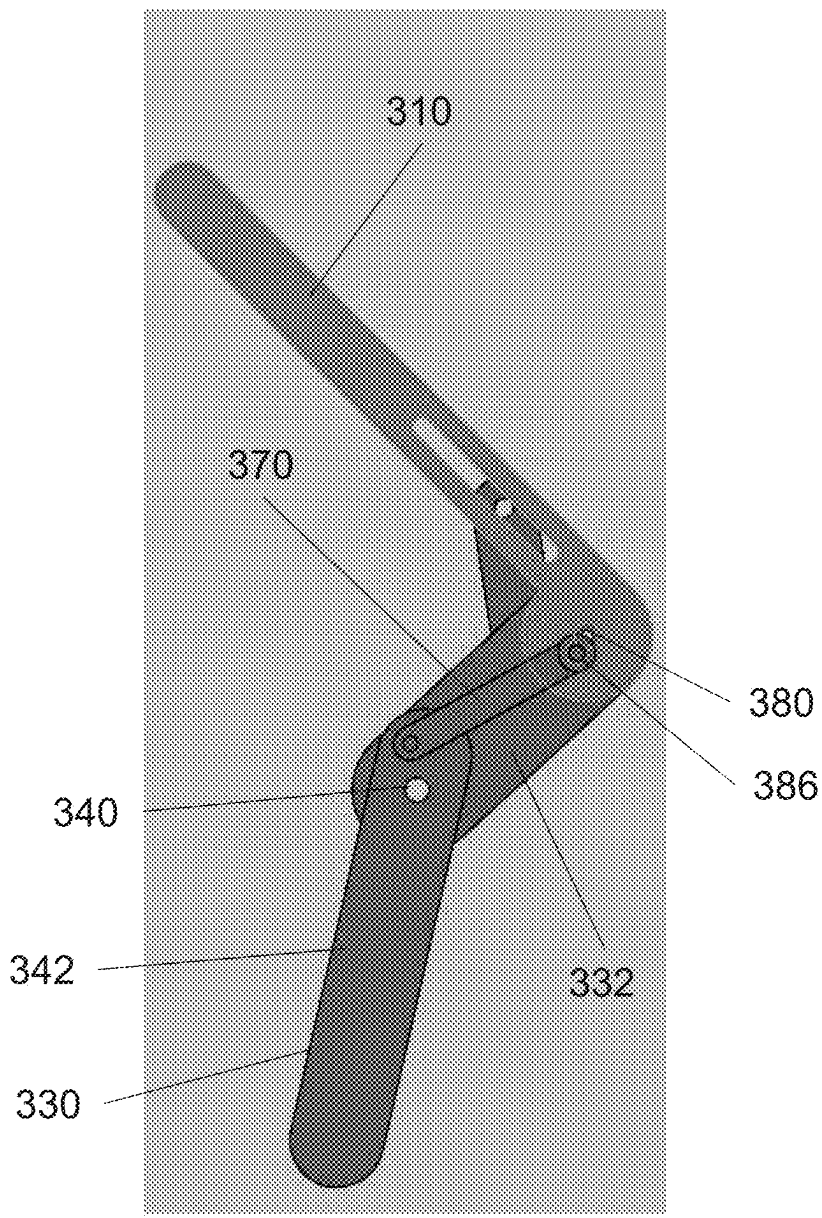

FIG. 7C

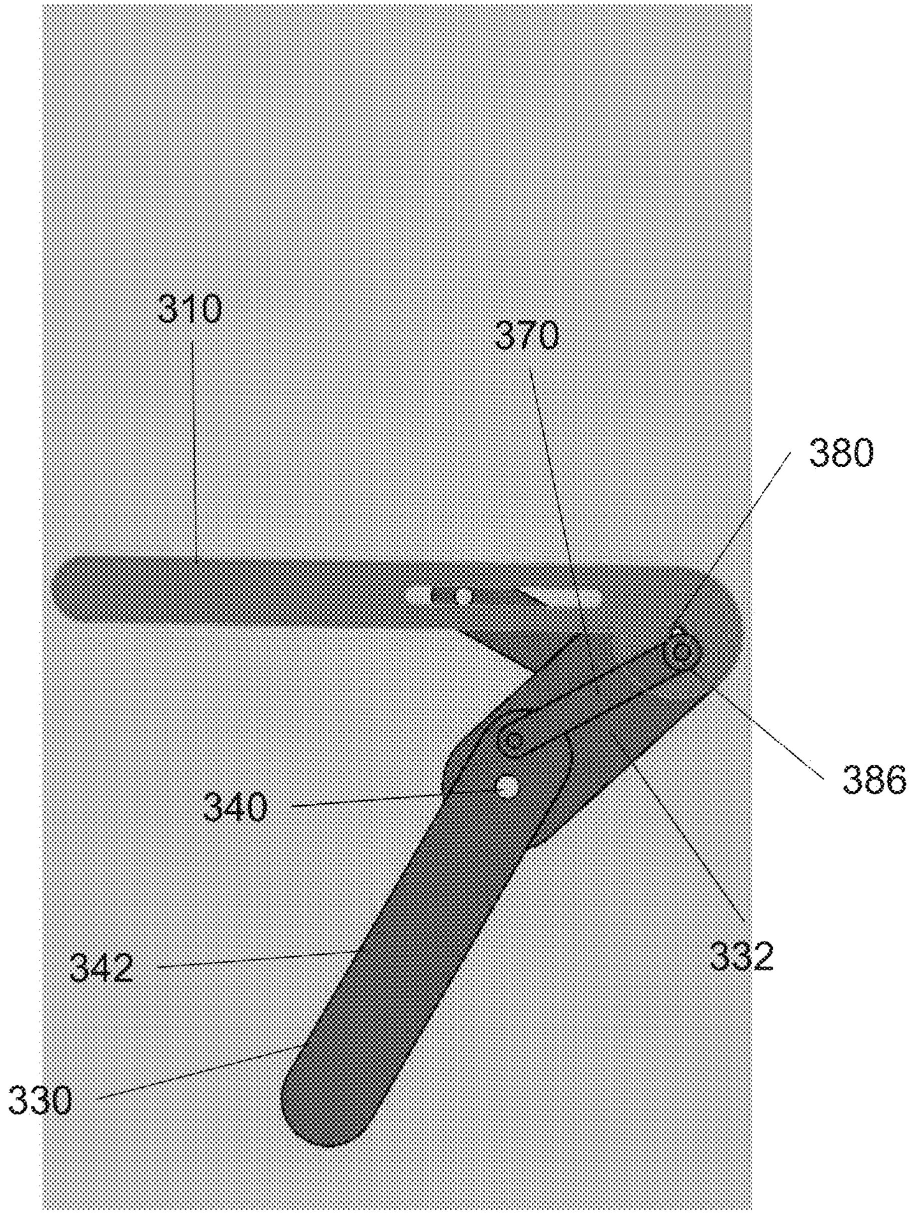

FIG. 7D

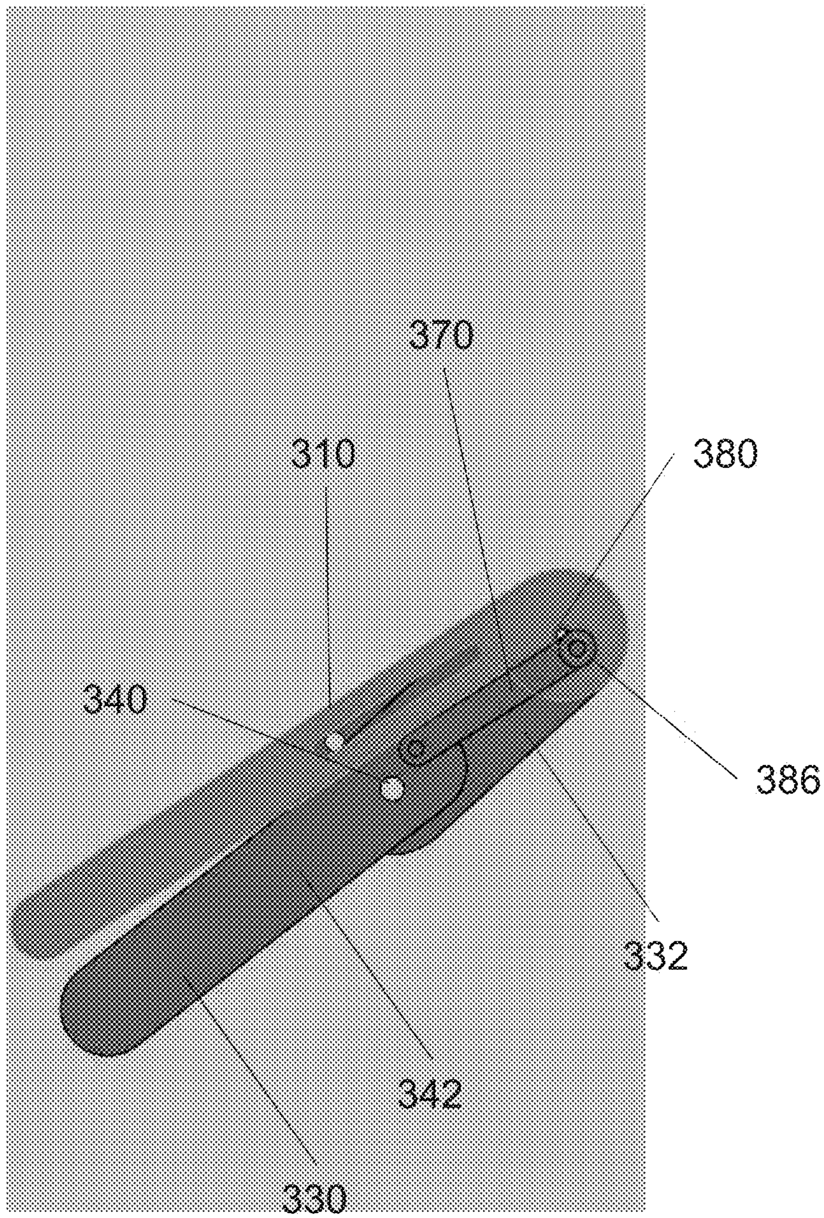

FIG. 7E

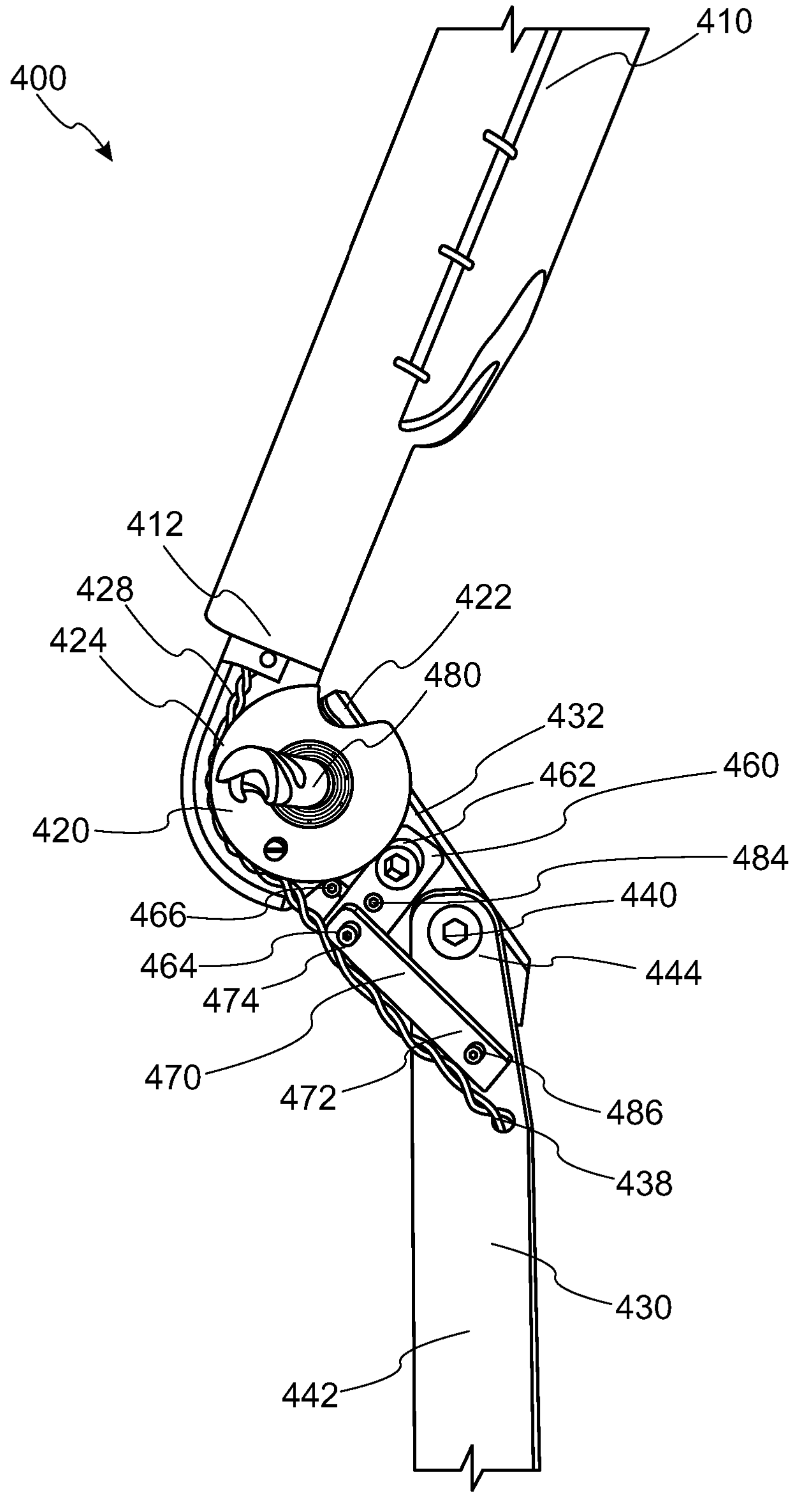
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FIG. 8A

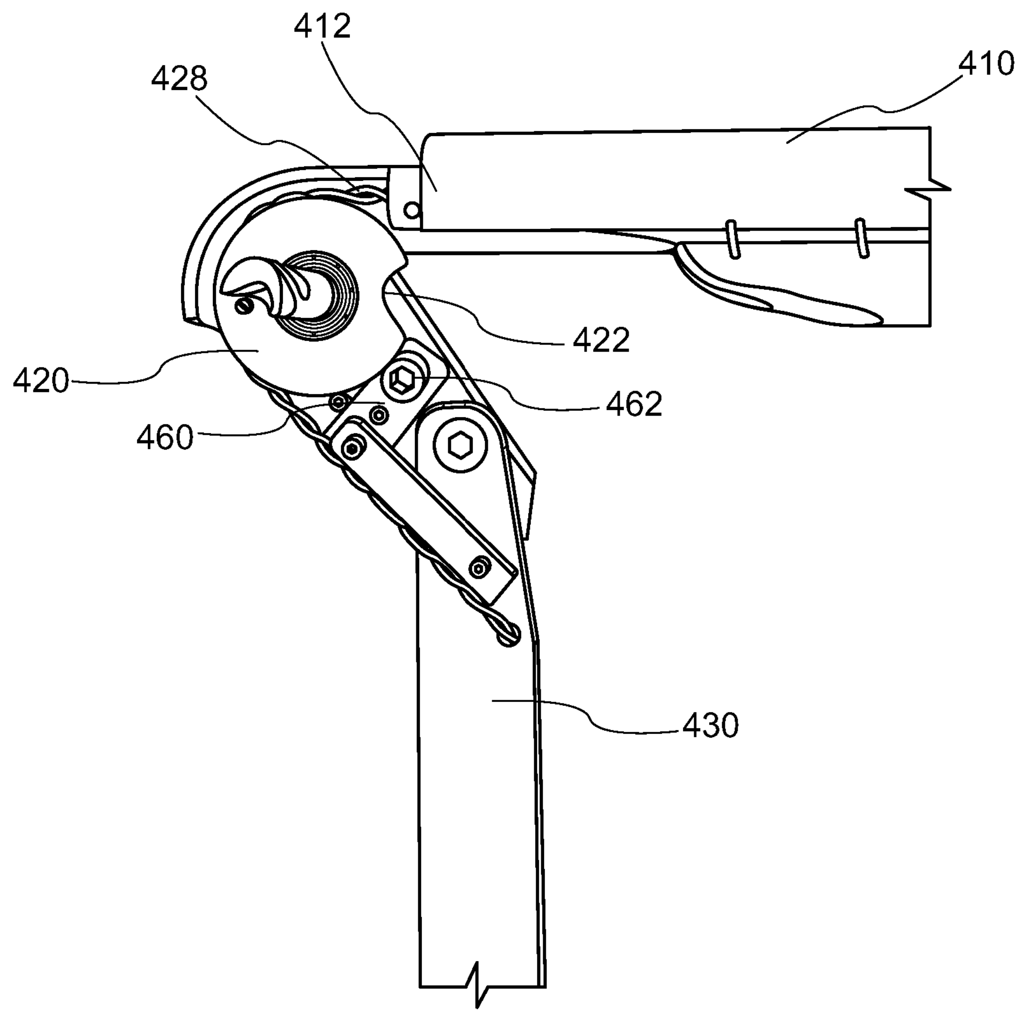

FIG. 8B

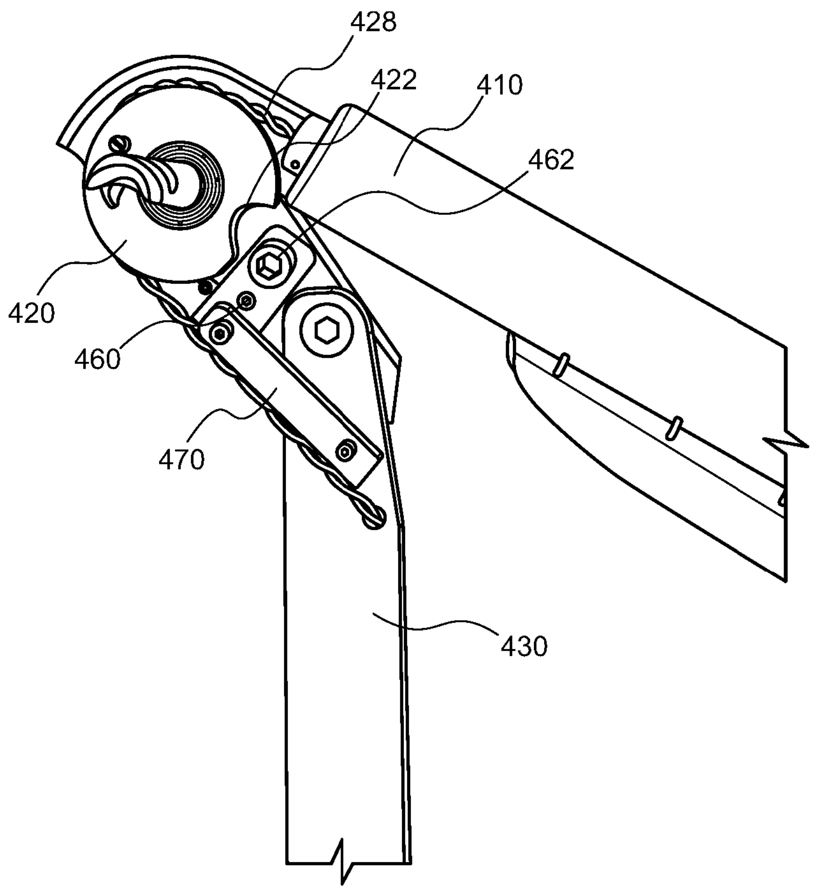

FIG. 8C

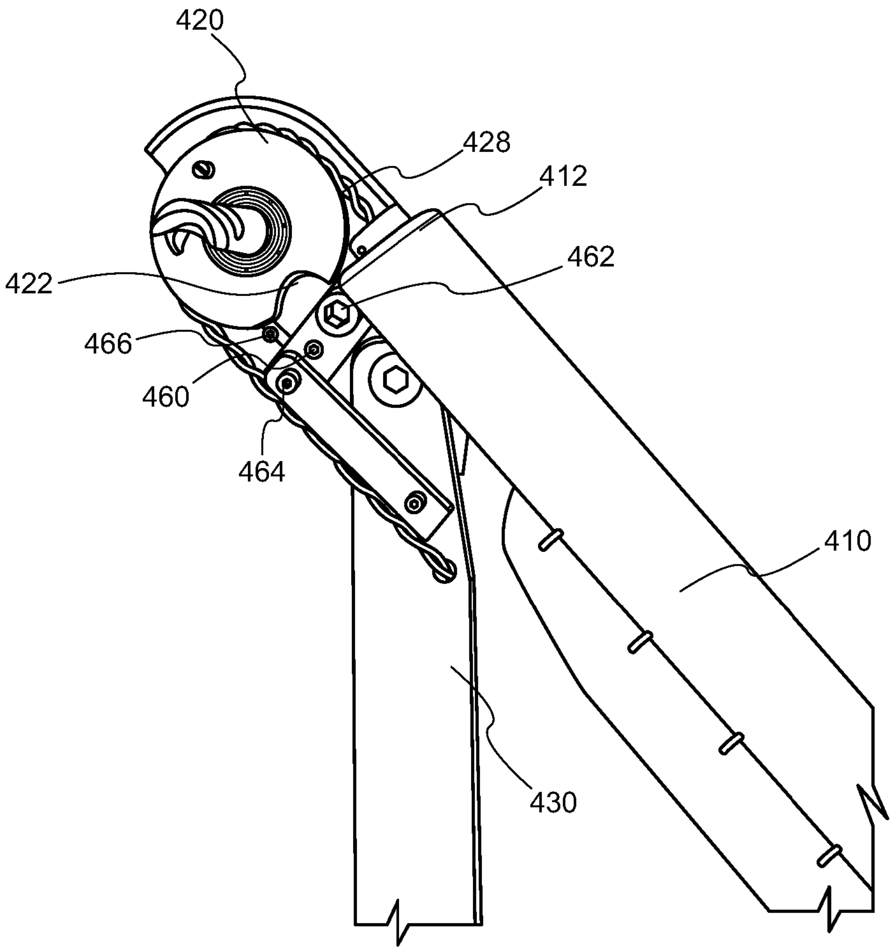

FIG. 8D

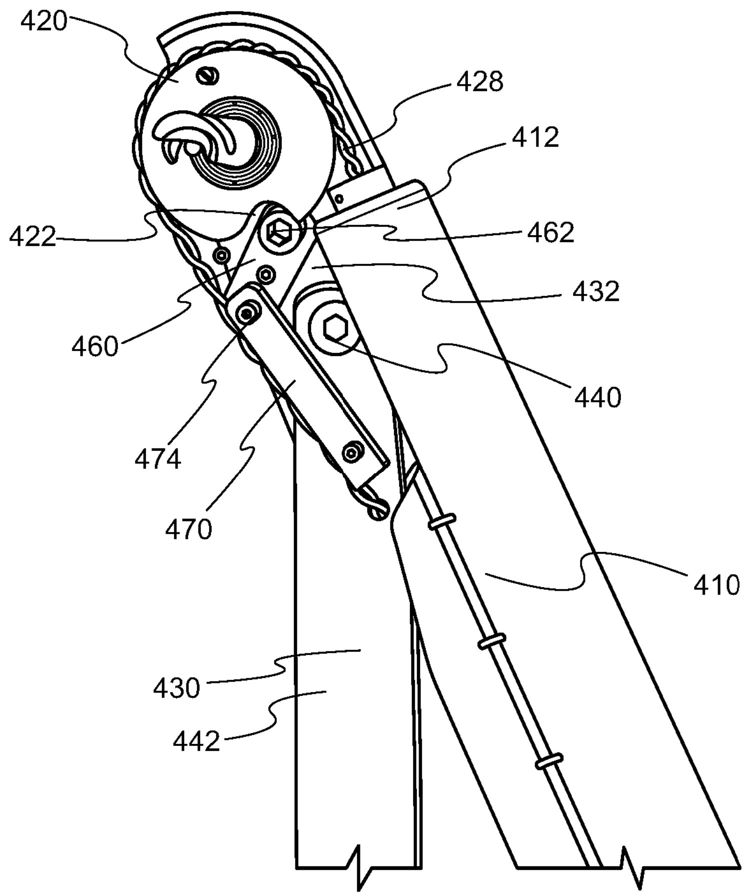

FIG. 8E

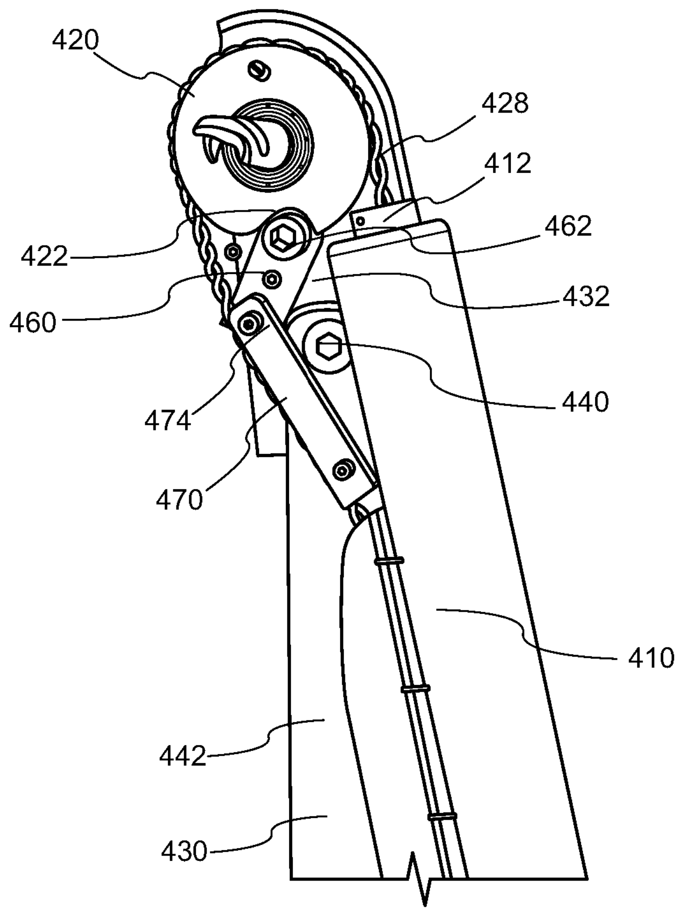

FIG. 8F

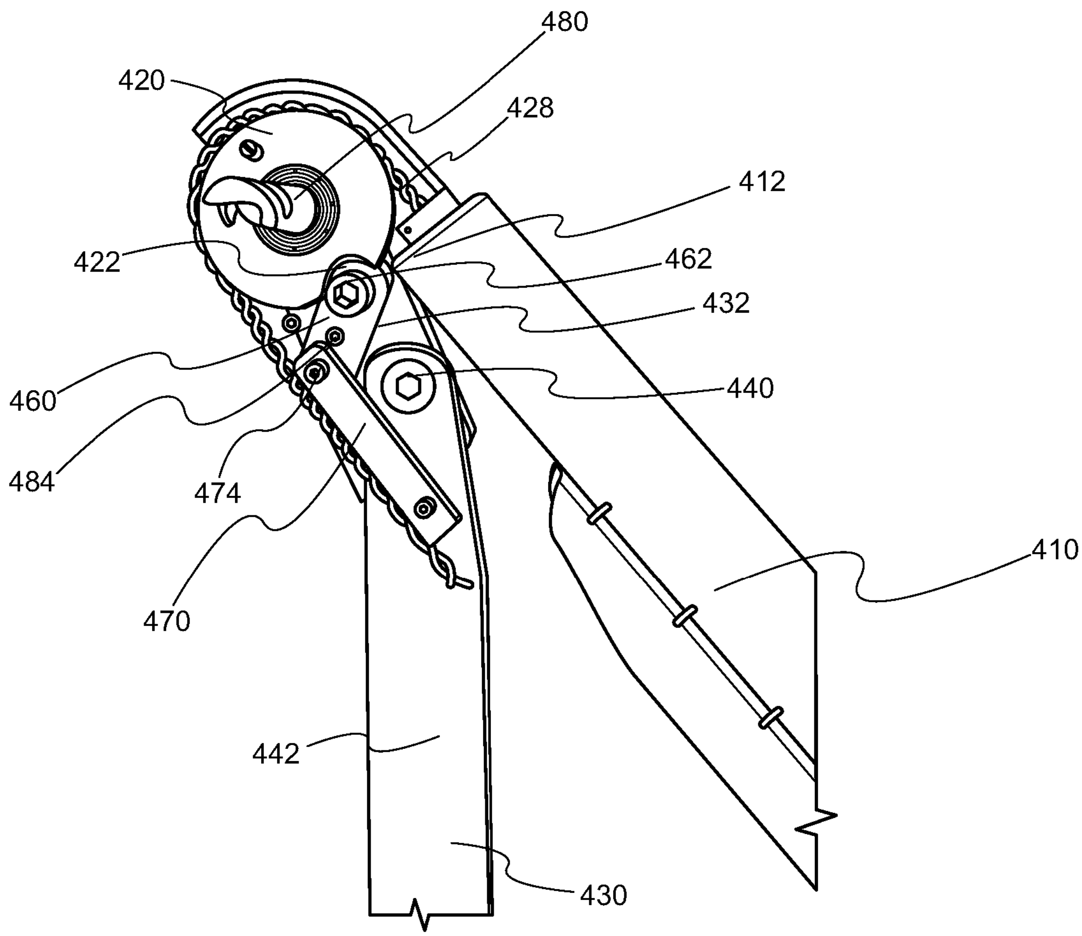

FIG. 8G

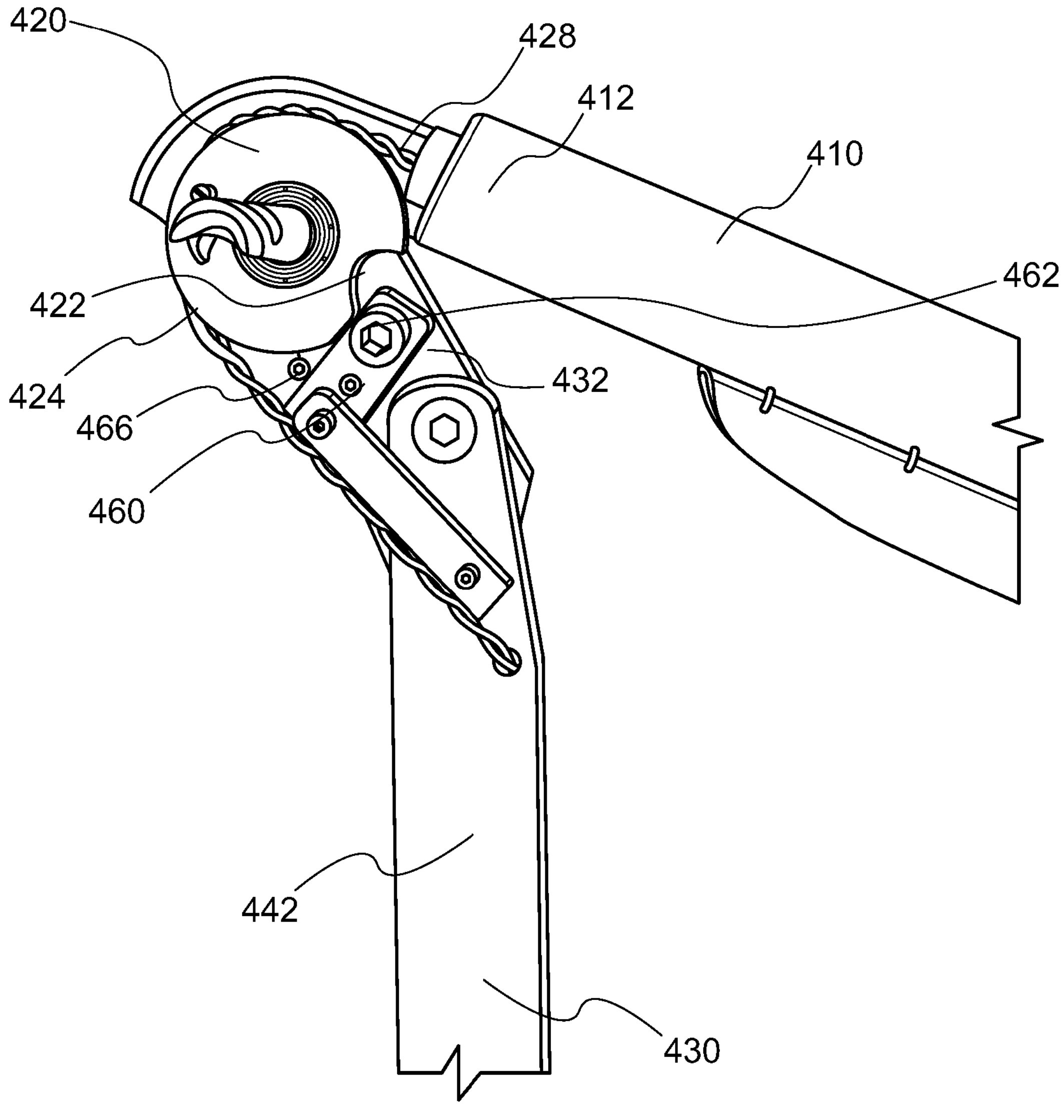

FIG. 8H

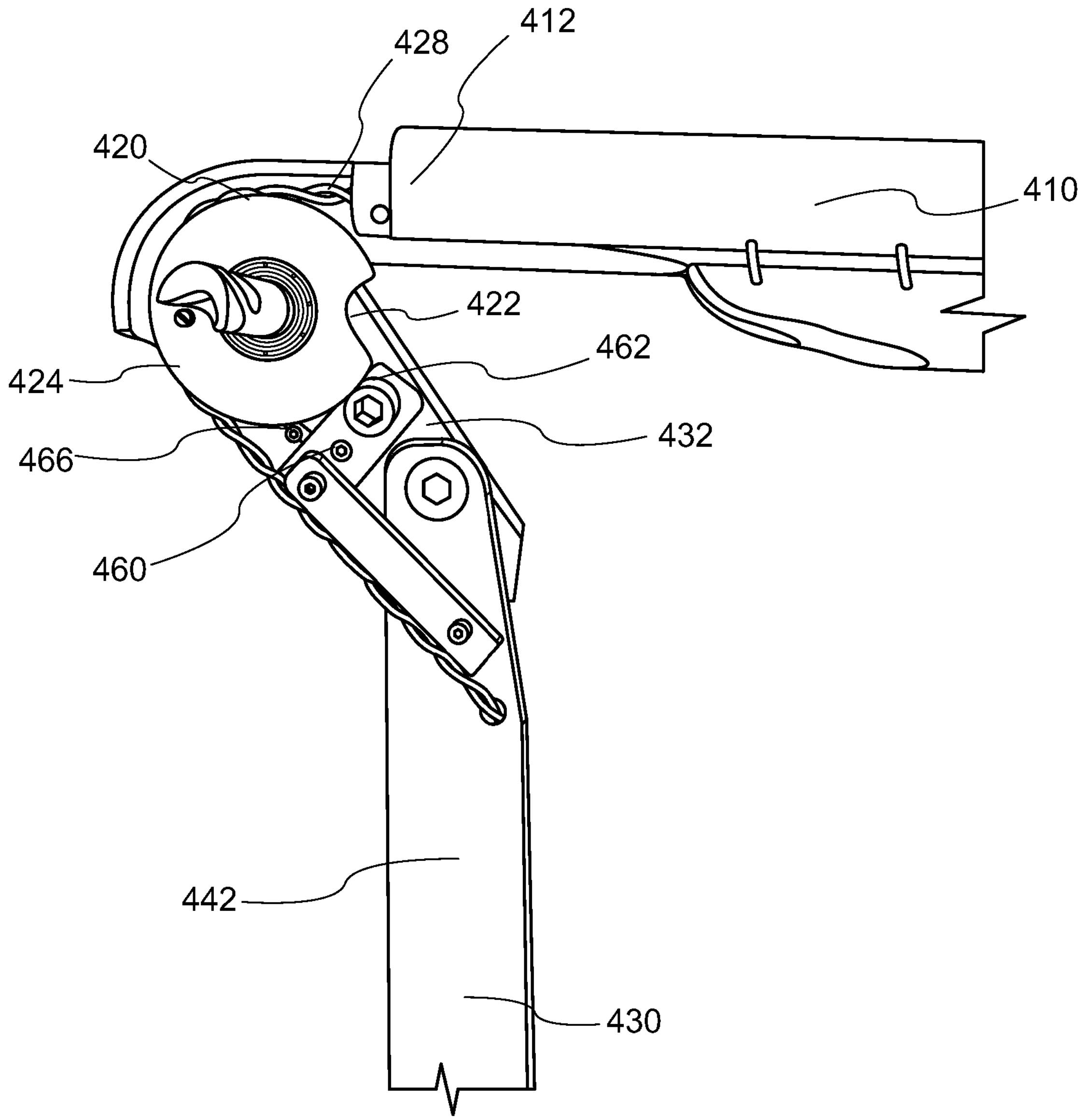

FIG. 8I

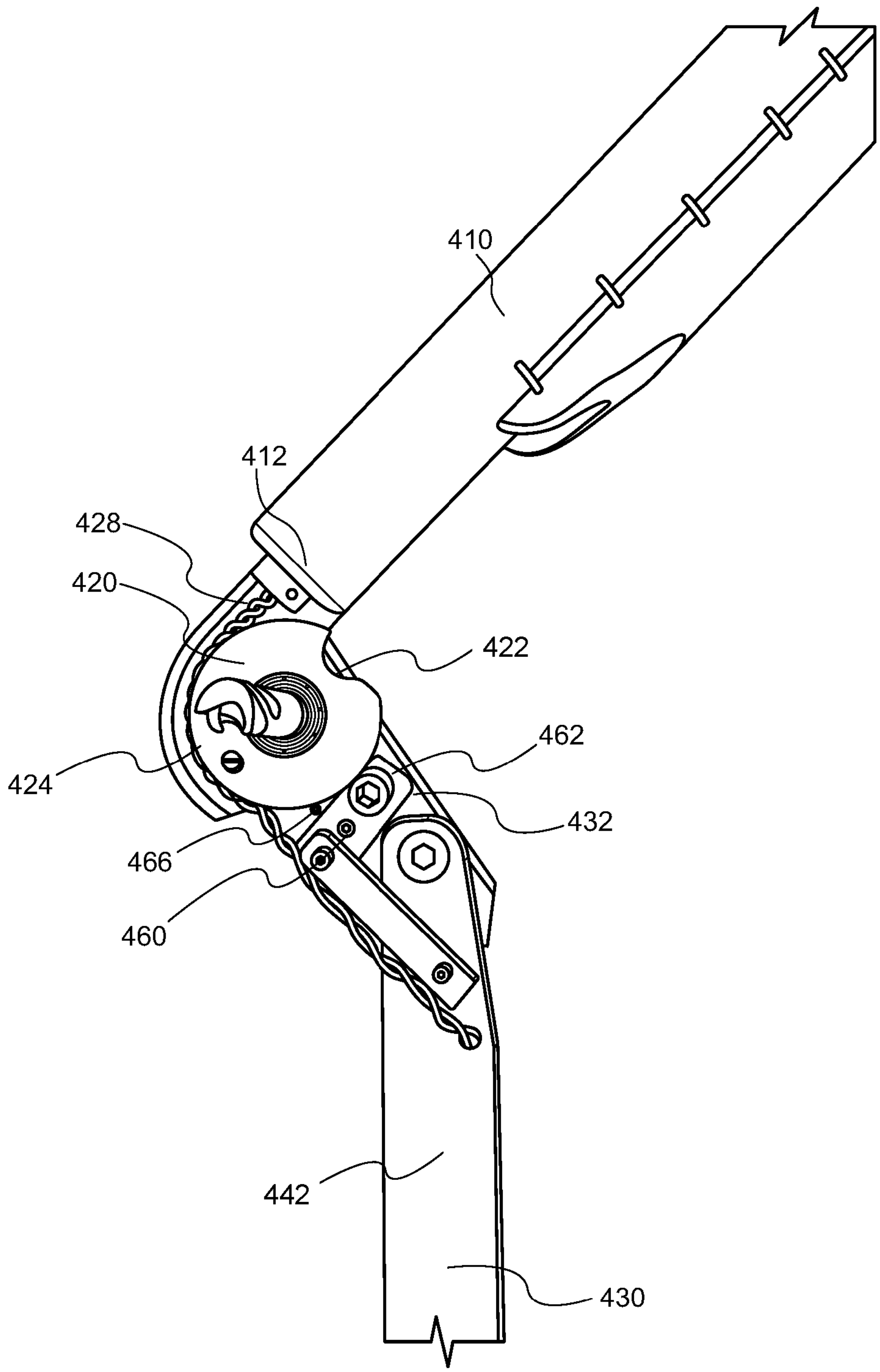

FIG. 8J

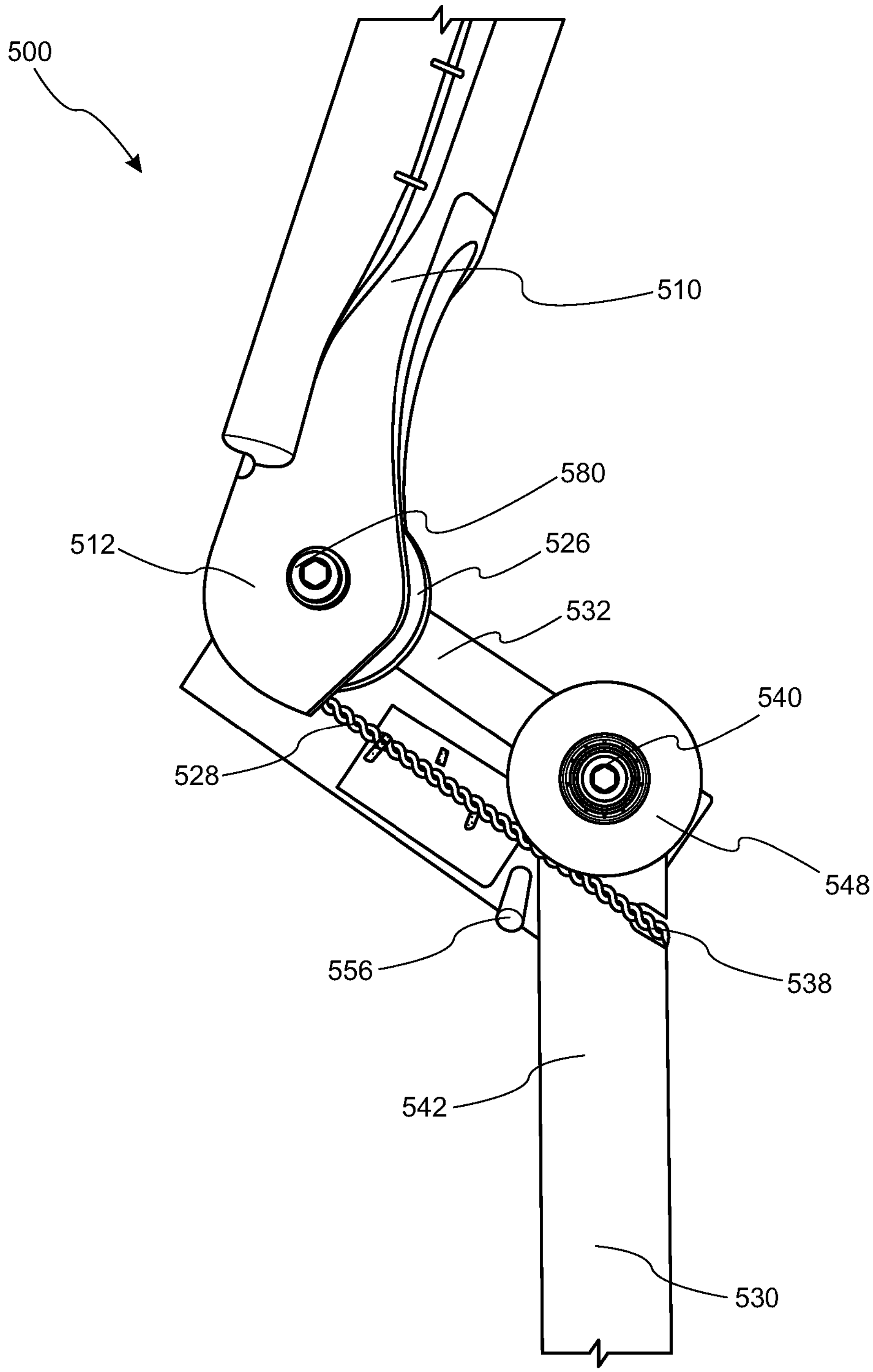

FIG. 9A

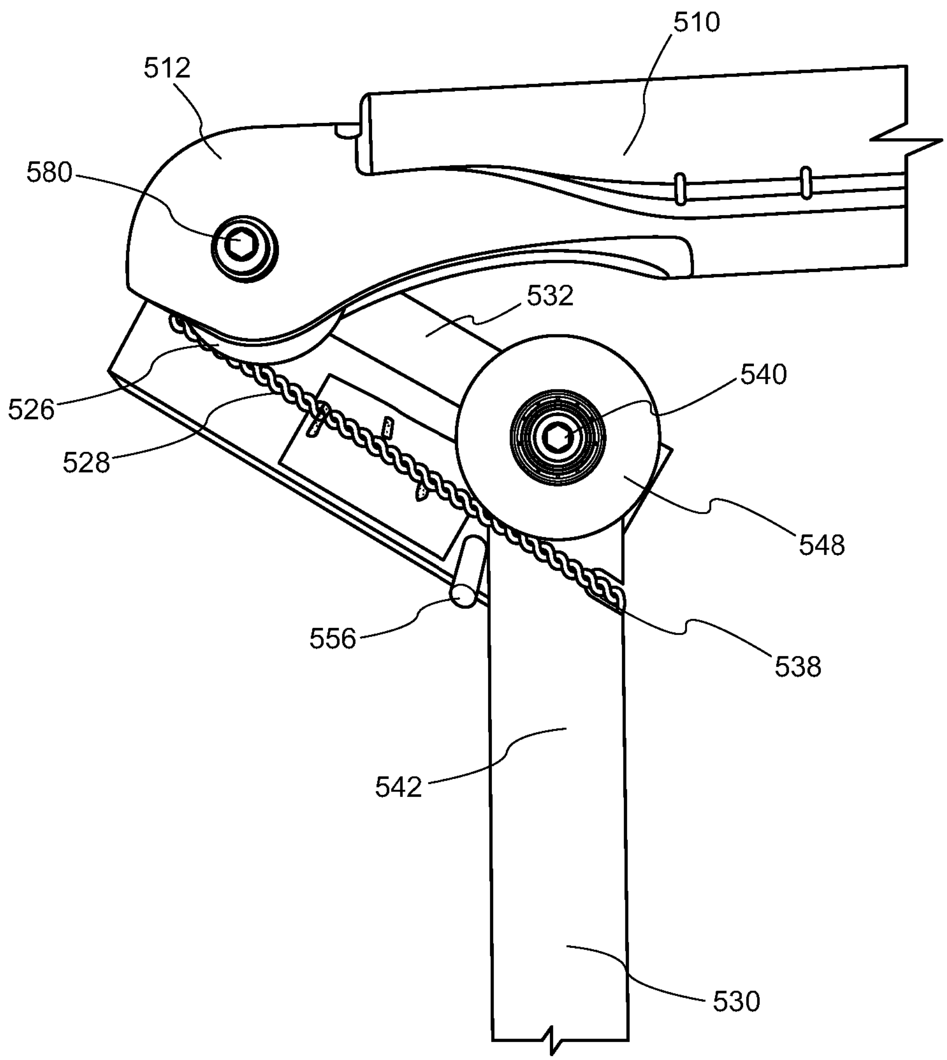

FIG. 9B

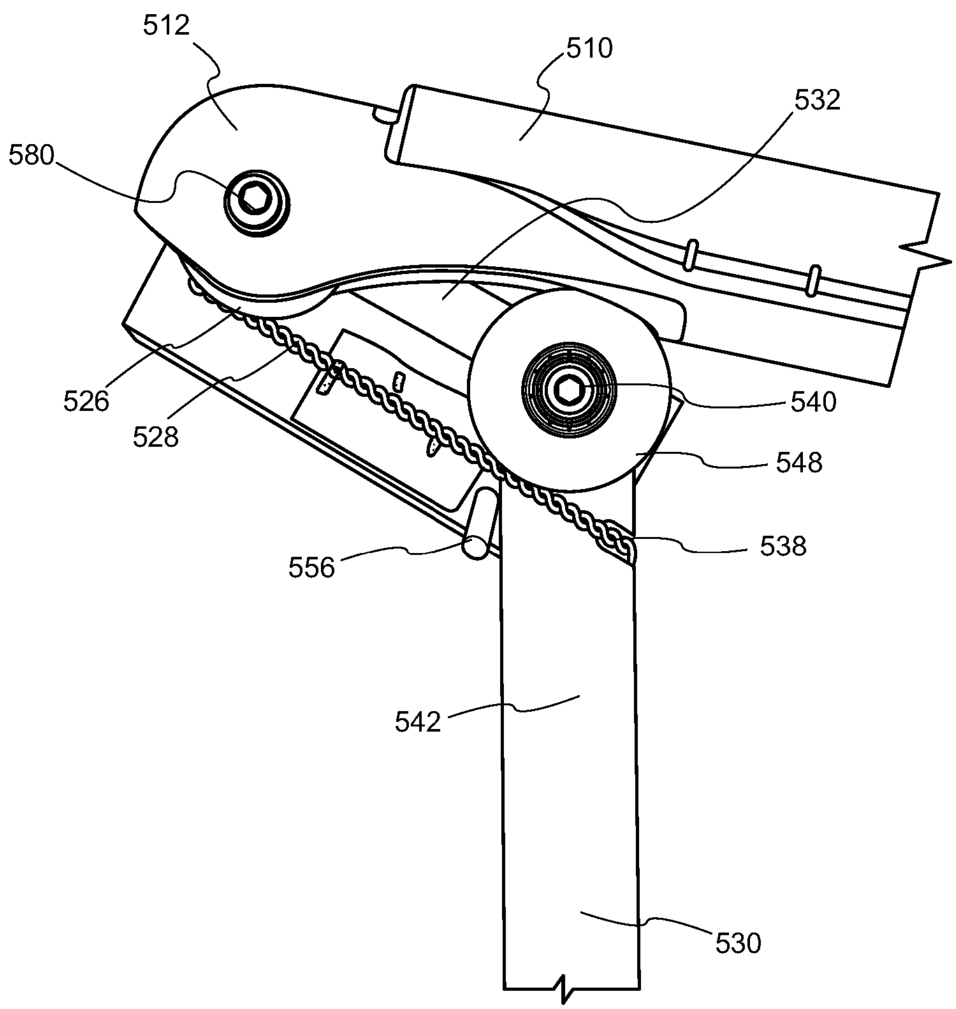

FIG. 9C

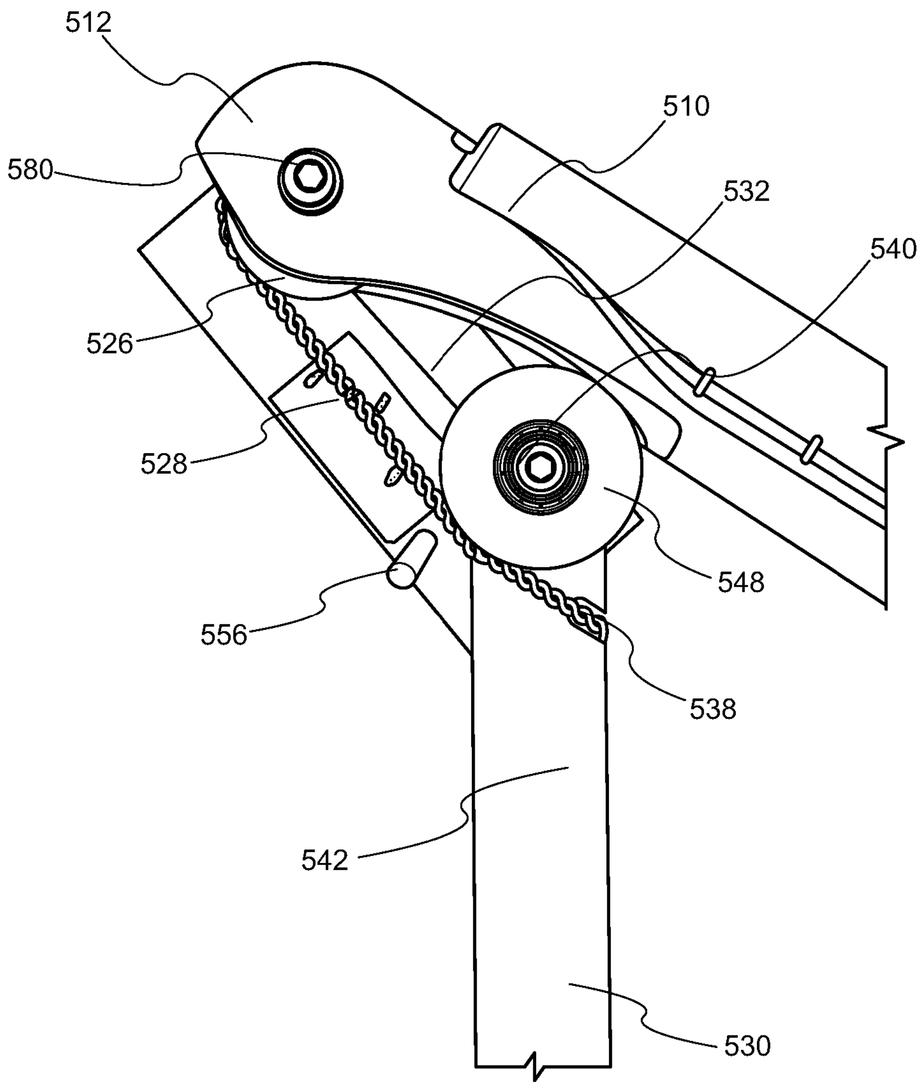

FIG. 9D

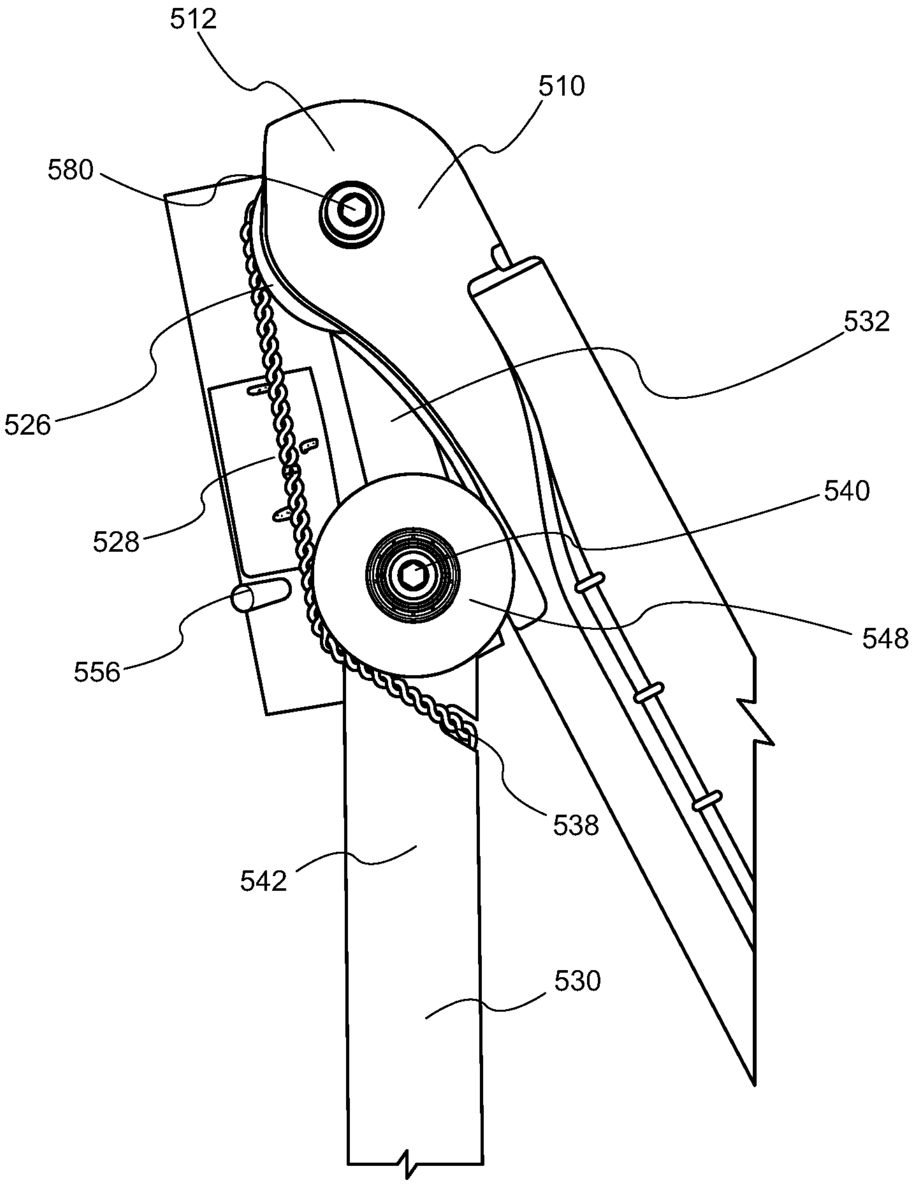

FIG. 9E

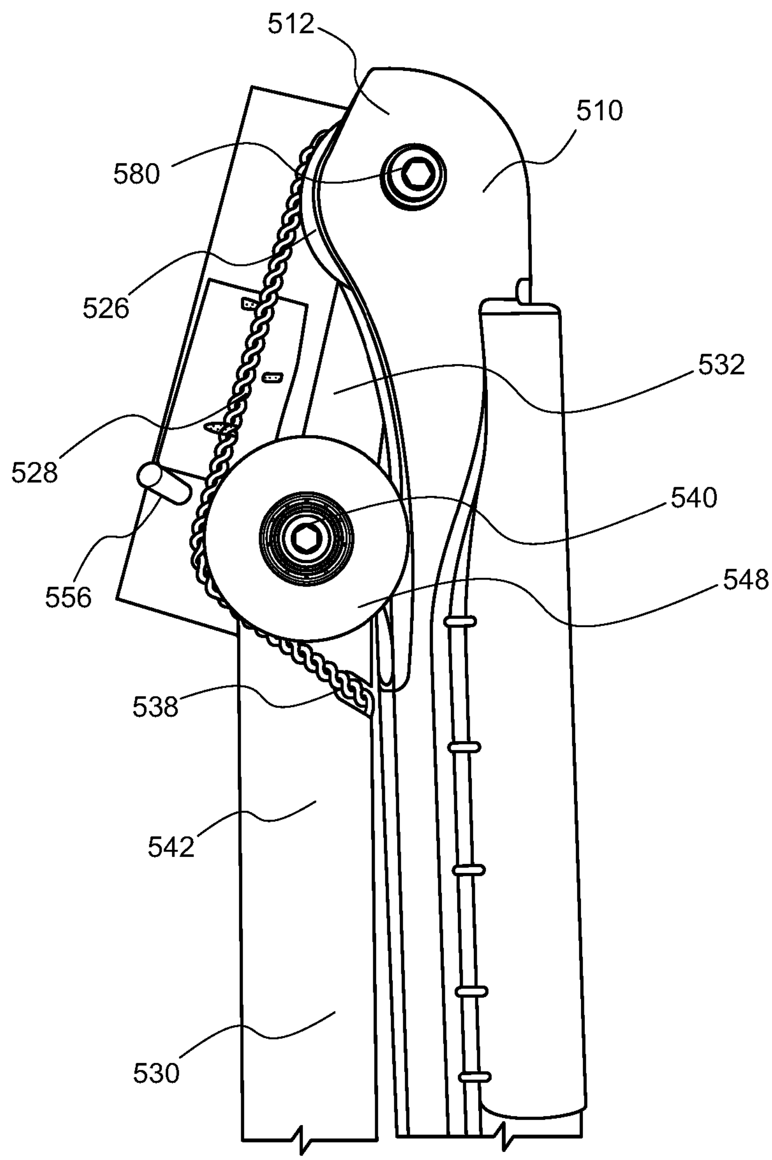

FIG. 9F

BODY WEIGHT SUPPORT

CROSS REFERENCES

[0001] This application is a U.S. National Phase Application of International Application No. PCT/GB2023/051268 filed May 15, 2023, which claims the priority of UK Patent Application No. GB2207217.7 filed May 17, 2022, the entire disclosures of each of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a body weight support. In particular, the present disclosure relates to a body weight support for supporting a user's body weight during an activity such as, but not limited to, skiing.

BACKGROUND

[0003] Skiing is a sport that can result in significant loads on the muscles and joints of the skier, particularly in the legs. The skier must repeatedly bend the legs, whilst at the same time the legs must continue to support the skier. The loads are often increased as a result of the skier adopting an incorrect stance or posture, or from repeated impacts of the skis with uneven compacted snow or ice, causing a bouncing effect. Consequently, all skiers can benefit from the use of body weight supports that assist in setting the skier in a correct position and from reducing the forces on the skier's leg muscles and joints.

[0004] Body weight supports can also be used in other settings to support a user's body weight and reduce the load on the user's muscles and joints. For example, body weight supports can be used to support a user's weight during manual operations such as fruit picking and assembly line work, or any other activity involving a squatting motion.

[0005] Existing body weight supports include an upper member configured for attachment to a user's upper leg (e.g. around the thigh), a lower member configured for attachment to a user's lower leg (e.g. by attachment to a boot worn on the user's foot), and a pivotal connection between the upper and lower members. Such existing body weight supports also include a load bearing element that resists the pivotal movement between the upper and lower members. The load bearing element reduces the load on the user's muscles and joints when the user bends their legs, thereby supporting the user in a squat position.

[0006] The upper member of existing body weight supports is configured for location against the side of the user's outer thigh, so that the pivotal connection between the upper and lower members is alongside the user's natural pivot point (i.e. the user's knee). The lower member of existing body weight supports is configured for attachment to the rear of the user's boot, in a substantially central position. Attachment of the body weight support to the rear of the user's boot prevents a twisting or turning force from being applied to the user's boot, which would occur, for example, if the lower member were attached to a side of the boot. Accordingly, such an attachment prevents the user's foot (and consequently the user's skis) from being turned inwards.

[0007] In order to accommodate the positioning of the upper member against the side of the user's outer thigh and the attachment of the lower member to the rear of the user's boot, existing body supports such as those described in EP 1066808B1 and available from Ski-Mojo Limited of Horsham, UK include a curved lower member. The curved lower member also provides a more ergonomic fit, by accommodating the user's calf muscles, which generally are not in line with the upper thigh and protrude laterally outwards. In particular, the lower member incorporates a curved section between the pivot and the point of attachment to the user's boot. A drawback of such solutions is that the curved lower member limits the angular rotation about the pivot, because the curved lower member is forced into contact with the upper member at acute angles. Such situations may occur, for example, if the user squats deeply, or during folding of the support when not in use. By forcing the lower member into contact with the upper member, the upper member and/or lower member can become damaged. A further drawback is that the support cannot be compactly collapsed when not in use, as the curved lower member protrudes outwardly.

[0008] Other existing body weight supports, such as those available from Againer of Riga, Latvia include an upper member in the form of an upper shell that houses the user's upper leg, and a lower member in the form of a lower shell that houses the user's lower leg. The upper and lower shell are pivotally coupled using pivots that are located, in use, on either side of the user's knee. A gas strut is connected between the upper shell and the lower shell, to resist the pivotal movement between the upper and lower shells. It will be appreciated that such body weight supports also fail to accommodate deep squatting or folding movements, because the piston can only be moved so far into the cylinder of the gas strut. The range of movement of the piston relative to the cylinder also prevents compact collapsing of the support when not in use.

[0009] Accordingly, there exists a need for an improved body weight support that addresses the drawbacks identified above.

SUMMARY

[0010] This summary introduces concepts that are described in more detail in the detailed description. It should not be used to identify essential features of the claimed subject matter, nor to limit the scope of the claimed subject matter.

[0011] According to one aspect of the present disclosure, there is provided a body weight support, comprising: a first member configured for location against a first portion of a limb of a user, the first portion of the limb being on a first side of a joint of the user; a second member configured for location against a second portion of the limb on a second side of the joint of the user, wherein the second member is rotatable relative to the first member about a pivot, the second member comprising: a first portion proximate to the pivot; and a second portion extending from the first portion, distal to the pivot; wherein the second portion is movable from a first configuration in which a first angle between the first portion and the first member is less than a second angle between the second portion and the first member, to a second configuration in which the difference between the first angle and the second angle is less than the difference between the first angle and the second angle in the first configuration; and a load bearing element configured to provide a resistance to rotation of the second member relative to the first member.

[0012] By allowing the difference between the first angle and the second angle to be reduced (i.e. in the second configuration of the second portion), the second portion of the second member can be moved to a position in which the angle between the second portion and the first member is small. In an example in which the first member is configured for location against an upper leg and the second member is configured for location against a lower leg, the reduction in angle between the second portion of the second member and the first member means that the body weight support can support the user at acute angles between the first member and second member, such as the angles resulting from deep squat positions. Moreover, the reduction in angle between the second portion of the second member and the first member means that the body weight support can be compactly stored when not in use.

[0013] The second member may be rotatable from a first angular position relative to the first member to a second angular position relative to the first member, wherein the second portion is not movable from the first configuration to the second configuration when the second member is in the first angular position, and wherein the second portion is movable from the first configuration to the second configuration when the second member is in the second angular position. The body weight support therefore implements a "delayed hinge", where the first configuration of the second portion is maintained until the second member is at a particular angular position relative to the first member.

[0014] The second member may be rotatable from a first angular position relative to the first member to a second angular position relative to the first member, wherein rotation of the second portion of the second member relative to the first member does not cause movement of the second portion from the first configuration to the second configuration when the second member is in the first angular position, and wherein rotation of the second portion of the second member relative to the first member causes movement of the second portion from the first configuration to the second configuration when the second member is in the second angular position. Such an arrangement also implements the "delayed hinge" described above.

[0015] The second member may be rotatable from a first angular position relative to the first member to a second angular position relative to the first member, wherein rotation of the second member from the first angular position to the second angular position causes movement of the second portion from the first configuration to the second configuration. The second portion therefore moves constantly relative to the first portion as the second member is rotated relative to the first member. When the second member is in the second angular position, an angle between the first portion and the first member may be smaller than when the second member is in the first angular position. In particular, when the second member is in the second angular position, the first portion may be substantially parallel to the first member.

[0016] When the second portion is in the second configuration, the second portion may be in line with the first portion. By providing a configuration in which the second portion is in line with the first portion, the user can be supported in deep squat positions and the body weight support can be compactly stored when not in use.

[0017] The second portion may be hingedly coupled to the first portion. The second portion may be prevented from hinging relative to the first portion until the second member is in the second angular position. Alternatively, the rotation of the second member relative to the first member may cause hinging of the second portion relative to the first portion.

[0018] The length of the second portion may be adjustable. Adjustment of the length of the second portion accommodates different lower limb lengths (e.g. different lower leg lengths). The length of the first portion may be adjustable. Adjustment of the length of the first portion accommodates different lower limb thicknesses (e.g. different lower leg thicknesses). An angle between the first portion and the second portion in the first configuration of the second member may be adjustable. The adjustment of the angle allows the body weight support to be adapted to the user's limbs, to optimise comfort.

[0019] The body weight support may further comprise a pivotal coupling configured to permit the rotation of the second member relative to the first member. The body weight support may further comprise a lever member coupled to the second member, wherein the lever member is configured to apply a frictional force to the pivotal coupling during rotation of the second member from the first angular position to the second angular position. The application of the frictional force to the pivotal coupling supplements the resistance provided by the load bearing element, to further resist rotation of the second member relative to the first member and thereby support the user's body weight.

[0020] The load bearing element may be deactivatable to permit substantially free rotation of the second member relative to the first member. Deactivation of the load bearing element allows the resistance provided by the load bearing element to be removed to allow for full leg mobility, thereby permitting normal sitting and walking movements while wearing the body weight support.

[0021] The first member may be configured for location against an upper leg of the user and the second member may be configured for location against a lower leg of the user. The first member may be configured for location against the outer side of the user's upper leg. This allows the pivot to be located adjacent to the user's knee, which is a natural pivot point. Accordingly, the second member pivots relative to the first member as the user's lower leg pivots relative to the user's upper leg.

[0022] The second portion may be connectable to a boot of the user. In particular, the second portion may be connectable to the rear of the boot of the user. Attachment of the body weight support to the rear of the user's boot prevents a twisting or turning force from being applied to the user's boot.

[0023] The body weight support may further comprise a connection member pivotally coupled to the first portion of the second member, wherein the connection member engages an end of the load bearing element. The length of the connection member may be adjustable. Adjustment of the length of the connection member allows the resistance provided by the load bearing element to be adjusted (e.g. depending on the user's weight). An end of the connection member may be slidably movable within a slot in the first member, and the end of the connection member may engage the end of the load bearing element.

[0024] The load bearing element may comprise a cable. Use of a cable allows a simpler mechanism to be provided to couple the second portion of the second member to the first member. In addition, the use of a cable reduces the force applied to a pivotal coupling that permits the rotation of the second member relative to the first member. Specifically, the force applied to the pivotal coupling is reduced when a first end of the cable is attached to the second portion of the second member. These and other aspects are merely illustrative of the innumerable aspects associated with the present disclosure and should not be deemed as limiting in any manner. These and other aspects, features, and advantages of the present disclosure will become apparent from the following detailed description when taken in conjunction with the referenced drawings.

BRIEF DESCRIPTION OF FIGURES

[0025] Reference is now made more particularly to the drawings, which illustrate the best presently known mode of carrying out the present disclosure and wherein similar reference characters indicate the same parts throughout the views.

[0026] FIG. **1** shows a front view of a body weight support according to a first embodiment, in which a lower member of the body weight support is shown at a first angular position with respect to an upper member of the body weight support.

[0027] FIG. **2** shows a simplified rear view of a body weight support shown in FIG. **1**, in which the lower member is shown at the first angular position.

[0028] FIGS. **3**A to **3**C show front views of the body weight support shown in FIG. **1**, in which the lower member is shown at various angular positions with respect to the upper member.

[0029] FIGS. **4**A to **4**C show simplified rear views of the body weight support shown in FIG. **1**, in which the lower member is shown at various angular positions corresponding to the angular positions shown in FIGS. **3**A to **3**C.

[0030] FIGS. **5**A to **5**D show front views of the body weight support shown in FIG. **1**, in which the lower member is shown at various angular positions with respect to the upper member, and in which a toggle clip of the body weight support is in an open position.

[0031] FIGS. **6**A to **6**D show simplified rear views of a body weight support according to a second embodiment, in which a lower member of the body weight support is shown at various angular positions with respect to an upper member of the body weight support.

[0032] FIGS. **7**A to **7**E show simplified rear views of a body weight support according to a third embodiment, in which a lower member of the body weight support is shown at various angular positions with respect to an upper member of the body weight support.

[0033] FIGS. **8**A to **8**J show schematic rear views of a body weight support according to a fourth embodiment, in which a lower member of the body weight support is shown at various angular positions with respect to an upper member of the body weight support.

[0034] FIGS. **9**A to **9**F show schematic rear views of a body weight support according to a fifth embodiment, in which a lower member of the body weight support is shown at various angular positions with respect to an upper member of the body weight support.

DETAILED DESCRIPTION

[0035] The following description of technology is merely exemplary in nature of the subject matter, manufacture and use of one or more inventions, and is not intended to limit the scope, application, or uses of any specific invention claimed in this application or in such other applications as may be filed claiming priority to this application, or patents issuing therefrom. The following definitions and non-limiting guidelines must be considered in reviewing the description of the technology set forth herein.

[0036] In the following detailed description numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. However, it will be understood by those skilled in the art that the present disclosure may be practiced without these specific details. For example, the present disclosure is not limited in scope to the particular type of industry application depicted in the figures. In other instances, well-known methods, procedures, and components have not been described in detail so as not to obscure the present disclosure.

[0037] The headings and sub-headings used herein are intended only for general organization of topics within the present disclosure and are not intended to limit the disclosure of the technology or any aspect thereof. In particular, subject matter disclosed in the "Background" may include novel technology and may not constitute a recitation of prior art. Subject matter disclosed in the "Summary" is not an exhaustive or complete disclosure of the entire scope of the technology or any embodiments thereof. Classification or discussion of a material within a section of this specification as having a particular utility is made for convenience, and no inference should be drawn that the material must necessarily or solely function in accordance with its classification herein when it is used in any given composition.

[0038] The citation of references herein does not constitute an admission that those references are prior art or have any relevance to the patentability of the technology disclosed herein. All references cited in the "Detailed Description" section of this specification are hereby incorporated by reference in their entirety.

[0039] Implementations of the present disclosure are explained below with particular reference to body weight supports that reduce the load on a user's leg muscles and joints. It will be appreciated, however, that the devices described herein may also be applied to body weight supports that reduce the load on other limbs (e.g. arms). Moreover, implementations of the present disclosure are explained below with particular reference to skiing. It will further be appreciated that the devices described herein are also suitable for other activities in which a user's body weight is to be supported in order to reduce the load on the user's muscles and joints.

[0040] FIG. **1** is a schematic diagram of a front view of a body weight support **100** according to a first embodiment, while FIG. **2** is a schematic diagram of a rear view of the body weight support **100**. FIG. **2** is a simplified view of the body weight support **100** shown in FIG. **1**, because the toggle clip **102** (described below) is not shown in FIG. **2**. The body weight support **100** comprises a first member in the form of an upper member **110** configured for location against an upper limb (e.g. an upper leg) of a user. The upper limb may be considered a first portion of a limb (e.g. a leg), which is on a first side of a joint (e.g. a knee) of the user. As one example, the upper member **110** may be secured against the outer side of the user's thigh using one or more straps (not shown in FIG. **1**). The upper member **110** comprises a first end **112** (best shown in FIG. **2**) that is fixedly coupled to a cam **120**, and a second end **114** opposite the first end **112**. The upper member **110** further comprises a first elongate slot in the form of a lower slot **116** extending along a portion of the upper member **110** towards the first end **112**, and a second elongate slot in the form of an upper slot **118** (best shown in FIG. **2**) extending along a portion of the upper member **110** towards the second end **114**.

[0041] As shown in FIG. **2**, the cam **120** to which the first end **112** of the upper member **110** is coupled comprises an indent **122** located adjacent to the point at which the cam **120** is coupled to the first end **112**. Aside from the indent **122**, the cam **120** comprises a surface **124** having a substantially constant radius.

[0042] Returning to FIG. **1**, the body weight support **100** further comprises a hinged second member in the form of a lower member **130**. The lower member **130** is configured for location against a lower limb (e.g. a lower leg) of the user. The lower limb may be considered a second portion of the limb (e.g. leg), which is on a second side of the user's joint. The lower member **130** comprises a first portion **132** having a first end **134** pivotally coupled to the centre of the cam **120** at a first pivot **180** (as best shown in FIG. **3**A), meaning that the first portion **132** is proximate to the first pivot **180**. The pivotal coupling between the first end **134** of the first portion **132** and the cam **120** allows the first portion **132** (and therefore the lower member **130**) to be rotated relative to the upper member **110**. Returning to FIG. **1**, it can be seen that the first portion **132** of the lower member **130** further comprises a second end **136** coupled to a hinge **140**.

[0043] The lower member **130** also comprises a second portion **142** having a first end **144** coupled to the hinge **140**, such that the second portion **142** extends distally from the first portion **132** and is distal to the first pivot **180**. The second portion **142** also has a second end **146** at its distal end. In one example, the second end **146** is configured for connection to a rear portion of a boot (not shown) worn by the user.

[0044] The lower member **130** is configured for location against the lower limb of the user such that the second portion **142** is substantially parallel to the user's lower limb. The first portion **132** sits against the user's lower limb and connects the second portion **142** to the cam **120**. In an example in which the body weight support **100** is worn on the user's leg, the second portion **142** is located against the rear of the user's lower leg, while the first portion **132** is located against the side of the user's lower leg, and extends between the second portion **142** and the cam **120** (which is adjacent to the user's knee joint). In one example, the length of the second portion **142** is adjustable (e.g. using a telescopic adjustment), in order to accommodate different lower limb lengths. In a further example, the length of the first portion **132** is adjustable (e.g. using a telescopic adjustment), in order to accommodate different lower limb thicknesses. In a further example, the angle between the first portion **132** and the second portion **142** is adjustable. For example, the second portion **142** may include a plurality of holes via which the first portion **132** may be hingedly attached to the second portion **142**. Each of the plurality of holes may be located a different distance from the second end **146** of the second portion **142**, thereby each allowing distinct connection angles between the first portion **132** and the second portion **142**.

[0045] The body weight support **100** further comprises a connection member **150** having a first end **152** that is slidably moveable within the lower slot **116** in the upper member **110**. A linear bearing (not shown) permits the first end **152** of the connection member **150** to slide within the lower slot **116**. A second end **154** of the connection member **150** is pivotally coupled to the first portion **132** of the lower member **130** at a second pivot **182**. The second pivot **182** allows the first end **152** of the connection member **150** to slide within the lower slot **116** as the lower member **130** is rotated relative to the upper member **110**. In the example shown in FIG. **1**, an anticlockwise rotation of the lower member **130** relative to the upper member **110** (i.e. towards the upper member **110**) causes the first end **152** of the connection member **150** to be slidably displaced away from the cam **120** (as shown, for example, in FIG. **3**A), whereas a clockwise rotation of the lower member **130** relative to the upper member **110** (i.e. away from the upper member **110**) causes the first end **152** of the connection member **150** to be slidably displaced towards the cam **120** (e.g. to the position shown in FIG. **1**).

[0046] As shown in FIG. **2**, the body weight support **100** further comprises a bogie **160** pivotally coupled to the first portion **132** of the lower member **130** at a third pivot **184** located between a first end **162** of the bogie **160** and a second end **164** (best shown in FIG. **1**) of the bogie **160**. In the configuration shown in FIGS. **1** and **2**, each of the first end **162** and the second end **164** contacts the substantially constant radius surface **124** of the cam **120**, with the first end **162** being disposed closer to the indent **122** than the second end **164**, as shown in FIG. **2**. Each end **162**, **164** comprises one or more bearings allowing the end **162**, **164** to move against the substantially constant radius surface **124** of the cam **120**.

[0047] A lever member in the form of a bogie lever **170** extends between the second portion **142** of the lower member **130** and the second end **164** of the bogie **160**. The bogie lever **170** has a first end **172** that is pivotally coupled to the second portion **142** of the lower member **130** at a fourth pivot **186** located between the first end **144** and the second end **146** of the second portion **142**. A second end **174** of the bogie lever **170** is pivotally coupled to the second end **164** of the bogie **160**. The pivotal connections of the bogie lever **170** mean that, in the configuration shown in FIG. **2**, a clockwise rotation of the second portion **142** of the lower member **130** urges the bogie **160** to rotate clockwise about the third pivot **184**, because the bogie lever **170** is under tension when the second portion **142** is rotated clockwise.

[0048] As shown in FIG. **1**, the body weight support **100** further comprises a toggle clip **102** that is pivotally coupled to the first end **114** of the upper member **110**. The toggle clip **102** is shown in a closed position in FIG. **1**. The body weight support **100** also comprises a spring engagement member **104** having a first end **106** that is slidably movable within the upper slot **118** in the upper member **110**. A linear bearing (not shown) permits the first end **106** of the spring engagement member **104** to slide within the upper slot **118**. A second end **108** of the spring engagement member **104** is pivotally coupled to the toggle clip **102** at a fifth pivot **188**. The fifth pivot **188** allows the first end **106** of the spring engagement member **104** to slide within the upper slot **118** as the toggle clip **102** is rotated from the closed position shown in FIG. **1** to the open position shown in FIGS. **5**A to **5**D. In the example shown in FIG. **1**, a clockwise rotation of the toggle clip **102** about the pivotal coupling at the first end **114** of the upper member **110** causes the first end **106** of the spring engagement member **104** to be slidably displaced away from the cam **120** (as shown, for example, in FIG. **5**A), while an anticlockwise rotation of the toggle clip **102** about the pivotal coupling at the first end **114** of the upper member

110 causes the first end **106** of the spring engagement member **104** to be slidably displaced towards the cam **120** (e.g. to the position shown in FIG. **1**).

[0049] A load bearing element (not shown) is coupled between the first end **106** of the spring engagement member **104** and the first end **152** of the connection member **150**. In one example, the load bearing element is coupled to the first end **152** of the connection member **150**, and is not fixedly coupled to the first end **106** of the spring engagement member **104**, such that the first end **106** of the spring engagement member **104** can be pulled away from the load bearing element when the toggle clip **102** is moved to the open position. It will be appreciated, however, that the load bearing element may instead be fixedly coupled to the first end **106** of the spring engagement member **104** and not fixedly coupled to the first end **152** of the connection member **150**. As a further alternative, the load bearing element may not be fixedly coupled to the first end **106** of the spring engagement member **104** or to the first end **152** of the connection member **150**, in which case the load bearing element may be held in a guide (e.g. a tube) between the first end **152** and the first end **106**, and engaged by the first end **152** and the first end **106** when the toggle clip **102** is moved to the closed position.

[0050] The load bearing element provides a resistance to rotation of the lower member **130** relative to the upper member **110**. In one example, the load bearing element is a spring that is compressed as the first end **152** of the connection member **150** moves towards the first end **106** of the spring engagement member **104**. The compression of the spring causes the spring to exert a force on the first end **152** of the connection member **150** to bias the first end **152** of the connection member **150** towards the cam **120**, thereby providing a resistance to rotation of the lower member **130** relative to the upper member **110**. Specifically, in the example shown in FIG. **1**, the spring provides a resistance to anticlockwise rotation of the lower member **130** relative to the upper member **110**. It will be appreciated that, in other examples, other types of load bearing element (e.g. one or more gas struts) may be implemented instead of a spring. In one example, the length of the connection member **150** is adjustable, in order to provide adjustable levels of support resulting from adjustment of the force exerted by the load bearing element when compressed. Accordingly, the force exerted by the load bearing element may be adjusted depending on the user's weight.

[0051] When the body weight support **100** is in the configuration shown in FIGS. **1** and **2**, the lower member **130** is in a first angular position relative to the upper member **110**. In the first angular position, the second portion **142** of the lower member **130** is in a first configuration relative to the first portion **132** of the lower member **130**. Specifically, the second portion **142** of the lower member **130** cannot be rotated relative to the first portion **132**. This is because the contact between the first and second ends **162**, **164** of the bogie **160** and the substantially constant radius surface **124** of the cam **120** prevents rotation of the bogie **160** about the third pivot **184** from being imparted via the bogie lever **170** by rotation of the second portion **142** either towards the upper member **110** or away from the upper member **110**. Instead, the contact between the ends **162**, **164** of the bogie and the substantially constant radius surface **124** results in a frictional force being applied to the cam **120** via the bogie lever **170** when the second portion **142** is rotated. The friction applied to the cam **120** provides additional resistance to rotation of the lower member **130** relative to the upper member **110**.

[0052] As shown in FIG. **1**, when the second portion **142** of the lower member **130** is in the first configuration relative to the first portion **132** of the lower member, the angle **190** between the first portion **132** and the upper member **110** is less than the angle **192** between the second portion **142** and the upper member **110**.

[0053] FIGS. **3**A to **3**C and **4**A to **4**C show rotation of the lower member **130** to various angular positions relative to the upper member **110**. As with FIG. **2**, FIGS. **4**A to **4**C show simplified views in which the toggle clip **102** and spring engagement member **104** are omitted. As shown in FIGS. **3**A and **4**A, further rotation of the lower member **130** relative to the upper member **110** results in the ends **162**, **164** of the bogie **160** traversing the substantially constant radius surface **124** such that the first end **162** of the bogie **160** approaches the indent **122**. FIGS. **3**A and **4**A also show that the rotation of the lower member **130** causes the first end **152** of the connection member **150** to be displaced away from the cam **120**. This means that the spring (or other load bearing element) is compressed between the first end **152** of the connection member **150** and the first end **106** of the spring engagement member **104**. The compression of the spring provides a resistance to the rotation of the lower member **130** towards the upper member **110**.

[0054] FIGS. **3**B and **4**B show the position of the lower member **130** when the first end **162** of the bogie **160** is about to move into the indent **122**. In this angular position, the second portion **142** of the lower member **130** is still in the first configuration. The rotation of the lower member **130** to the position shown in FIGS. **3**B and **4**B causes the first end **152** of the connection member **150** to be displaced yet further from the cam **120**, further compressing the spring such that the resistance to rotation of the lower member **130** towards the upper member **110** is increased when compared to the angular position shown in FIGS. **3**A and **4**A.

[0055] As shown in FIGS. **3**C and **4**C, once the first end **162** of the bogie **160** reaches the indent **122**, the lower member **130** is in a second angular position relative to the first member **110**. In particular, when the lower member **130** is in the second angular position, the angle **190** between the first portion **132** and the upper member **110** is less than when the lower member **130** is in the first angular position. Specifically, the first portion **132** is substantially parallel to the upper member **110** (such that the angle **190** between the first portion **132** and the upper member **110** is close to zero degrees). At the second angular position, the first end **162** of the bogie **160** can move in a radially inward direction, towards the first pivot **180**. This means that the bogie **160** can rotate about the third pivot **184**.

[0056] The rotation of the bogie **160** permits the second portion **142** of the first member **130** to be moved out of the first configuration shown in FIGS. **3**B and **4**B. In particular, the second portion **142** can be rotated relative to the first portion **132** about the hinge **140**, so that the second portion **142** can be moved to a second configuration relative to the first portion **132**. In the example shown in FIGS. **3**C and **4**C, when the second portion **142** is in the second configuration, the second portion **142** is aligned with (i.e. in line with) the first portion **132**.

[0057] It will be appreciated that when the second portion **142** is in the second configuration relative to the first portion

132, the difference between the angle **190** and the angle **192** (close to zero in the example shown in FIGS. **3**C and **4**C) is less than the difference between the angle **190** and the angle **192** when the second portion **142** is in the first configuration relative to the first portion **132**. The rotation of the lower member **130** to the position shown in FIGS. **3**C and **4**C causes the first end **152** of the connection member **150** to be displaced even further from the cam **120**, further compressing the spring such that the resistance to rotation of the lower member **130** towards the upper member **110** is increased when compared to the angular position shown in FIGS. **3**B and **4**B.

[0058] FIGS. **5**A to **5**D show rotation of the lower member **130** to angular positions that correspond to the angular positions shown in FIG. **1** and FIGS. **3**A to **3**C, but with the toggle clip **102** in an open position. As explained above, the load bearing element (e.g. a spring) is coupled to the first end **152** of the connection member **150**, but is not fixedly coupled to the first end **106** of the spring engagement member **104**. Accordingly, when the lower member **130** is in the first angular position relative to the upper member **110**, shown in FIG. **5**A, the first end **106** of the spring engagement member **104** is free of the end of the spring.

[0059] In one example, the length of the spring is approximately equal to the distance between the first end **106** of the spring engagement member **104** and the first end **152** of the connection member **150** when the toggle clip **102** is in the closed position and when the lower member **130** is in the first angular position relative to the upper member **110** (e.g. as shown in FIG. **1**). This means that when the toggle clip **102** is in the open position and the lower member **130** is rotated towards the upper member **110** (e.g. as shown in FIGS. **5**B and **5**C), the spring is not compressed. This is because the distance between the first end **152** and the first end **106** in the configurations shown in FIGS. **5**B and **5**C exceeds the distance between the first end **152** and the first end **106** in the configuration shown in FIG. **1**. Accordingly, the user can deactivate the load bearing element so that they can adopt a seated or squatting position without substantial resistance to rotation of the lower member **130** (i.e. such that there is substantially free rotation of the second member **130** relative to the first member **110**). In this context, the term "substantially free rotation" is intended to convey that there will be a small amount of resistance to rotation resulting from the friction of the bogie **160** against the cam **120**, but that no specific load bearing element is activated.

[0060] As shown in FIG. **5**D, the body weight support **100** can be compactly collapsed, without compression of the spring, by rotating the lower member **130** to the second angular position relative to the upper member **110** with the toggle clip **102** in the open position. This means that the body weight support **100** can be easily folded, as the user does not have to overcome the resistance provided by the spring when folding the body weight support **100** with the toggle clip **102** in the open position.

[0061] The body weight support **100** described above provides the functionality to allow for location of the upper member **110** against the outer side of the user's upper leg while permitting connection of the lower member **130** to the rear of the user's boot, thereby reducing the loads on the user's muscles and joints without imparting an inward twisting force on the user's foot. This functionality is provided by the hinged lower member **130**, particularly when the second portion **142** of the lower member **130** is in the first configuration. The body weight support **100** described above also provides the functionality to support the user's body weight when in a deep squat position (i.e. with a small angle between the upper member **110** and the lower member **130**). This functionality is provided by the movement of the second portion **142** of the lower member **130** out of the first configuration when the lower member **130** is rotated sufficiently far towards the upper member **110**. The ability to move the second portion **142** of the lower member **130** to the second configuration also means that the body weight support **100** can be compactly folded.

[0062] Variations or modifications to the systems and methods described herein are set out in the following paragraphs.

[0063] In the embodiment shown in FIGS. **1** to **5**D, rotation of the bogie **160** is imparted by rotation of the lower member **130** towards the upper member **110**. The rotation of the lower member **130** (specifically, the second portion **142** of the lower member **130**) applies a tensile force to the bogie lever **170**. This tensile force urges the bogie **160** to rotate, but the rotation of the bogie **160** is prevented until the lower member **130** is rotated to the angular position in which the first end **162** of the bogie **160** can move into the indent **122**.

[0064] In an alternative embodiment, the first end **144** of the lower portion **142** can extend beyond the hinge **140**, the first end **172** of the bogie lever **170** can be pivotally coupled to the first end **144** of the lower portion **142**, and the second end **174** of the bogie lever **170** can be pivotally coupled to the first end **162** of the bogie **160**. In such a configuration, the rotation of the second portion **142** applies a compressive force to the bogie lever **170**. This compressive force on the first end **162** of the bogie **160** urges the bogie **160** to rotate, but the rotation of the bogie **160** is prevented until the lower member **130** is in the second angular position. In this position, the compressive force on the first end **162** of the bogie **160** causes the first end **162** of the bogie **160** to be pushed into the indent **122**, thereby causing rotation of the bogie **160**.

[0065] FIGS. **6**A to **6**D show an alternative pivotal connection of a body weight support **200** according to a second embodiment. As with the first embodiment, a first member in the form of an upper member **210** has a first end **212** that is fixedly coupled to a cam **220** having an indent **222** and a substantially constant radius surface **224**.

[0066] A second member in the form of a lower member **230** comprises a first portion **232** and a second portion **242**. The first portion **232** of the lower member **230** is pivotally coupled to the cam **220** about a first pivot **280** and pivotally coupled to the second portion **242** of the lower member **230** at a hinge **240**.

[0067] The first portion **232** of the lower member **230** also includes a slot **294**. The slot **294** comprises an annular portion **294***a* that has an outer radius that matches the radius of the substantially constant radius surface **224** of the cam **220**, and is coincident with a portion of the cam surface. The slot **294** also includes a radial portion **294***b* that extends radially outwards from the end of the annular portion **294***a* that is closest to the indent **222**. It will be appreciated that as the slot **294** is in the first portion **232** of the lower member **230**, the slot **294**, first pivot **280** and hinge **240** are in a fixed relationship with respect to one another.

[0068] The pivotal connection also includes a bearing **296** that is movable within the slot **294** such that the bearing **296** can move into the annular portion **294***a* and into the radial portion **294***b* of the slot **294** (depending on the angular position of the lower member **230** with respect to the upper member **210**, as described below). The bearing **296** is configured to rotate as the lower member **230** is rotated relative to the upper member **210**.

[0069] The body weight support **200** also includes a lever member **270** having a first end **272** that is pivotally coupled to the second portion **242** of the lower member **230**. In particular, the first end **272** is pivotally coupled to the second portion **242** at a fourth pivot **286** located between the first end **244** and the second end **246** of the second portion **242**. The second end **274** of the lever member is pivotally coupled to the bearing **296**. When the second portion **242** of the lower member **230** is rotated, a tensile force is applied to the lever member **270**. The tensile force via the lever member **270** pulls the bearing **296** against the slot **294**.

[0070] During rotation of the lower member **230** from the angular position shown in FIG. **6**A to the angular position shown in FIG. **6**B, the bearing **296** rotates within the radial portion **294***b* of the slot **294**. Once the lower member **230** is rotated to the angular position shown in FIG. **6**C, any further rotation of the lower member **230** towards the upper member **210** allows the bearing **296** to move radially inwards into the indent **222** in the surface of the cam **220**. Once the bearing **296** has moved radially inwards into the indent **222**, the bearing **296** can move within the annular portion **294***a* of the slot **294**. Rotation of the lower member **230** towards the upper member **210** applies a tensile force to the lever member **270**, which pulls the bearing **296** to the end of the annular portion **294***a* that is closest to the hinge **240** and causes the second portion **242** of the lower member **230** to move to its second configuration.

[0071] The pivotal coupling according to the second embodiment results in reduced friction force on the cam **220** compared to the friction exerted on the cam **120** of the body weight support **100** of the first embodiment. This is because some of the force applied via the lever member **270** is transferred to the slot **294** in the first portion **232** of the lower member **230**, instead of the cam **220**. The reduced friction on the cam **220** reduces the wear on the cam **220**, which may prolong the life of the cam **220**.

[0072] FIGS. **7**A to **7**E show a further alternative pivotal connection of a body weight support **300** according to a third embodiment. As with the first embodiment, a first member in the form of an upper member **310** has a first end **312** that is fixedly coupled to a cam **320**. However, unlike the first and second embodiments, the cam **320** of the third embodiment has a substantially constant radius surface **324** and no indent.

[0073] A second member in the form of a lower member **330** comprises a first portion **332** and a second portion **342**. The first portion **332** has a first end **334** pivotally coupled to the cam **320** at a first pivot **380**, and a second end **336** coupled to a hinge **340**. The second portion **342** has a first end **344** that extends beyond the hinge **340**. This means that the hinge **340** is located towards the first end **344** of the second portion **342**, but between the first end **344** and the second end **346**.

[0074] The body weight support **300** also includes a lever member **370** having a first end **372** and a second end **374**. The second end **374** of the lever member **370** is pivotally coupled to the cam **320** at a third pivot **384**. The first end **372** of the lever member **370** is pivotally coupled to the first end **344** of the second portion **342** at a fourth pivot **386**.

[0075] The third pivot **384** and the fourth pivot **386** are located such that the angle between the lever member **370** and the upper member **310** is smaller than the angle between the first portion **332** and the upper member **310**. In addition, the third pivot **384** is radially offset from the first pivot **380** so that when the lower member **330** is at the first angular position, the third pivot **384** is closer to the hinge **340** than the first pivot **380**.

[0076] Unlike the body weight supports **100**, **200** of the first and second embodiments, which only permit the second portion of the lower member to be moved from the first configuration to the second configuration when the lower member is at a particular angular position relative to the upper member, the second portion **342** of the lower member **330** of the body weight support **300** of the third embodiment moves constantly as the lower member **330** is rotated relative to the upper member **310**. This is because the rotation of the lower member **330** relative to the upper member **310** causes rotation of the second portion **342** of the lower member **330** from the first configuration (FIG. **7**A) to the second configuration (FIG. **7**E).

[0077] Specifically, as shown in FIGS. **7**A to **7**E, the radial offset of the third pivot **384** from the first pivot **380** means that the third pivot **384** rotates about the first pivot **380** as the lower member **330** is rotated relative to the upper member **310**. The rotation of the third pivot **384** about the first pivot **380** results in increased distance between the third pivot **384** and the hinge **340** as the lower member **330** is rotated. The increase in distance between the third pivot **384** and the hinge **340** allows the second portion **342** of the lower member **330** to be gradually straightened relative to the first portion **332** of the lower member **330** as the lower member **330** is rotated relative to the upper member **310**.

[0078] The body weight supports **200**, **300** of the second and third embodiments provide resistance to rotation of the lower member relative to the upper member, in the same way as for the body weight support **100** of the first embodiment. In addition, such resistance can be deactivated using a toggle clip in the same manner as for the body weight support **100** of the first embodiment.

[0079] FIGS. **8**A to **8**J show a fourth embodiment of a body weight support **400**, in the form of a modified version of the first embodiment described with reference to FIGS. **1** to **5**D. As with the first embodiment, a first member in the form of an upper member **410** has a first end **412** that is fixedly coupled to a cam **420** having an indent **422** and a substantially constant radius surface **424**.

[0080] A hinged second member in the form of a lower member **430** comprises a first portion **432** and a second portion **442**. The first portion **432** of the lower member **430** is pivotally coupled to the cam **420** about a first pivot **480** and pivotally coupled to the second portion **442** of the lower member **430** at a hinge **440**.

[0081] As with the first embodiment, the body weight support **400** also comprises a bogie **460** pivotally coupled to the first portion **432** of the lower member **430** at a second pivot **484** located between a first end **462** of the bogie **460** and a second end **464** of the bogie **460**. However, unlike the first embodiment, the second end **464** of the bogie **460** of the fourth embodiment does not contact the substantially constant radius surface **424** of the cam **420**. Instead, the second end **464** of the bogie **460** is in contact with a stop **466** that protrudes from the first portion **432** of the lower member **430**. The stop **466** prevents the second end **464** of the bogie

460 from contacting the cam **420** when the bogie **460** is rotated about the second pivot **484**. Consequently, the stop **466** reduces the friction force applied by the bogie **460** to the cam **420**.

[0082] The body weight support **400** also includes a lever member in the form of a bogie lever **470**, which extends between the second portion **442** of the lower member **430** and the second end **464** of the bogie **460**. The bogie lever **470** has a first end **472** that is pivotally coupled to the second portion **442** of the lower member **430** at a third pivot **486** located between a first end **444** and a second end (not shown) of the second portion **442**. A second end **474** of the bogie lever **470** is pivotally coupled to the second end **464** of the bogie **460**.

[0083] In contrast to the first embodiment, the body weight support **400** of the fourth embodiment does not include a connection member that is pivotally coupled to the first portion **432** of the lower member **430**. Instead, the body weight support **400** includes a load bearing element comprising a replaceable cable **428**. A first end **438** of the cable **428** is attached to the second portion **442** of the lower member **430** adjacent to the third pivot **486**. Specifically, the first end **438** of the cable **428** is attached to the second portion **442** at a position such that the distance between the first end **438** of the cable **428** and the hinge **440** is greater than the distance between the third pivot **486** and the hinge **440**. In other words, the first end **438** of the cable **428** is attached to the second portion **442** at a point that is more distal than the third pivot **486**. Preferably, the tangential distance between the hinge **440** and the cable **428** is slightly greater than the radius of a pulley (described further below) that is coincident with the cam **420**. This ensures full rotation of the lower member **430** (specifically, the first portion **432** of the lower member **430**) relative to the upper member **410** prior to rotation of the second portion **442** of the lower member **430** about the hinge **440**.

[0084] The load bearing element also includes a spring (not shown), which is housed within the upper member **410**. A second end (not shown) of the cable **428** is attached to the end of the spring that is furthest from the cam **420**, meaning that the cable **428** protrudes from an opening in the first end **412** of the upper member **410**. The spring is compressed when the cable **428** is pulled out of the first end **412** (i.e. as the lower member **430** is rotated relative to the upper member **410** from the angular position shown in FIG. **8**A to the angular position shown in FIG. **8**C, for example).

[0085] The cable **428** is in contact with a pulley (not shown) that is coincident with the cam **420**. While the cam **420** is fixedly connected to the upper member **410**, the pulley is free to rotate under the action of the cable **428** as the lower member **430** is rotated relative to the upper member **410**, such that the cable **428** does not slide over the pulley. Although not shown in FIG. **8**A, a sandwich construction may be used, in which the pulley is disposed between two symmetrical cams **420**. The pulley is offset from the cam **420** in the axial direction (i.e. into the drawing shown in FIG. **8**A), so that it does not interfere with the rotation of the first end **462** of the bogie **460** into the indent **422**. The pulley radius may be substantially equal to the radius of the substantially constant radius surface **424** of the cam **420**.

[0086] As described above, the bogie lever **170** of the body weight support **100** of the first embodiment is under tension when the second portion **142** of the lower member **130** is rotated towards the upper member **110**. The tension applied to the bogie lever **170** urges the first end **162** of the bogie **160** to rotate towards the first pivot **180**, which results in a frictional force being applied to the cam **120** by the first end **162** of the bogie **160**. In contrast, the tensile force applied by the cable **428** of the body weight support **400** of the fourth embodiment means that the tension applied to the bogie lever **470** is reduced or even eliminated. Consequently, the first end **462** of the bogie **460** is not urged to rotate towards the first pivot **480**, meaning that the frictional force applied to the cam **420** by the first end **462** of the bogie **460** is reduced or even eliminated. This reduces the frictional force on the cam **420**, thereby reducing wear on the cam **420**.

[0087] In some examples, the force applied by the cable **428** may be sufficiently high that a compressive force is applied to the bogie lever **470**, thereby urging the first end **462** of the bogie **460** to rotate away from the first pivot **480**. However, as explained above, rotation of the bogie **460** in this manner is prevented by the abutment of the second end **464** of the bogie **460** against the stop **466**. FIG. **8**A shows the lower member **430** in a first angular position relative to the upper member **410**. Rotation of the lower member **430** anticlockwise relative to the upper member **410** (in the orientation shown in FIG. **8**A) causes movement of the lower member **430** to the position shown in FIG. **8**B, in which the first end **462** of the bogie **460** is adjacent to the indent **422** in the cam **420**.

[0088] When moving from the position shown in FIG. **8**A to the position shown in FIG. **8**B, the contact between the cable **428** and the pulley means that the cable **428** is pulled out of the first end **412** of the upper member **410** as the lower member **430** is rotated towards the upper member **410**. Therefore, the length of cable **428** protruding from the first end **412** of the upper member **410** in the position shown in FIG. **8**B exceeds the length of cable **428** protruding from the first end **412** in the position shown in FIG. **8**A, meaning that the spring within the upper member **410** is compressed as the lower member **430** is moved towards the upper member **410**. The compression of the spring resists the rotational movement of the lower member **430** from the angular position relative to the upper member **410** shown in FIG. **8**A to the angular position relative to the upper member **410** shown in FIG. **8**B. In other words, the compression of the spring caused by drawing the cable **428** out of the first end **412** of the upper member **410** resists the rotational movement of the lower member **430** towards the upper member **410**.

[0089] As shown in FIG. **8**C, further rotation of the lower member **430** relative to the upper member **410** does not result in the first end **462** of the bogie **460** being rotated into the indent **422** in the cam **420** (unlike the body weight support **100** of the first embodiment). This is because the tensile force applied by the cable **428** eliminates the tension in the bogie lever **470**, meaning that the first end **462** of the bogie **460** is not being urged to rotate into the indent **422**.

[0090] Instead, the lower member **430** is rotated to the position shown in FIG. **8**D, in which the first end **462** of the bogie **460** contacts the first end **412** of the upper member **410**. The first end **412** of the upper member **410** is fixedly connected to the cam **420** at an angular position that is immediately adjacent to the indent **422**, meaning that the contact between the first end **412** of the upper member **410** and the first end **462** of the bogie **460** forces the first end **462** of the bogie **460** into the indent **422** (as shown in FIG. **8**E).

Rotation of the first end **462** of the bogie **460** in the opposite direction (i.e. away from the indent **422**) is prevented by the stop **466**, which abuts the second end **464** of the bogie **460**. As mentioned above, this means that no force from the second end **464** of the bogie **460** is applied to the cam **420**.

[0091] In the position shown in FIG. **8**E, the first end **462** of the bogie **460** has been forced into the indent **422** by the action of the first end **412** of the upper member **410**. This means that the second end **474** of the bogie lever **470** is rotated towards the hinge **440**, thereby allowing the second portion **442** of the lower member **430** to be rotated relative to the first portion **432** of the lower member **430** (i.e. moved from a first configuration shown in FIGS. **8**A to **8**D to a second configuration shown in FIG. **8**F). The rotation of the second portion **442** of the lower member **430** relative to the first portion **432** of the lower member **430** is as described above for the body weight support **100** of the first embodiment. As will be appreciated, rotation of the lower member **430** from the angular position relative to the upper member **410** shown in FIG. **8**B to the angular position relative to the upper member **410** shown in FIG. **8**E draws additional cable **428** out of the first end **412** of the upper member **410**, further compressing the spring, resulting in further resistance to such rotation.

[0092] As shown in FIGS. **8**D to **8**F, the tangential distance between the cable **428** and the hinge **440** reduces as the second portion **442** of lower member **430** is rotated relative to the first portion **432** of the lower member **430** from the angular position shown in FIG. **8**D to the angular position shown in FIG. **8**E. The tangential distance between the cable **428** and the hinge **440** is further reduced when the second portion **442** is moved from the angular position relative to the first position **432** shown in FIG. **8**E to the relative angular position shown in FIG. **8**F.

[0093] The reduction in the tangential distance between the cable **428** and the hinge **440** means that when the lower member **430** is rotated in the opposite direction (i.e. away from the upper member **410**, by rotation from the angular position shown in FIG. **8**F to the angular position shown in FIG. **8**G), the first portion **432** of the lower member **430** pivots relative to the upper member **410** about the first pivot **480** before the second portion **442** of the lower member **430** hinges relative to the first portion **432** of the lower member **430** about the hinge **440**. This is because the moment about the first pivot **480** exceeds the moment about the hinge **440** (as a result of the radius of the pulley exceeding the tangential distance between the cable **428** and the hinge **440**).

[0094] When the lower member **430** is moved away from the upper member **410** by angular rotation in the opposite direction (i.e. from the angular position shown in FIG. **8**F to the angular position shown in FIG. **8**G), the indent **422** forces the first end **462** of the bogie **460** to rotate away from the first pivot **480**. The surface of the indent **422** located furthest from the attachment point of the first end **412** of the upper member **410** may be curved in order to provide a smooth transition of the first end **462** of the bogie **460** out of the indent **422**, as shown in FIGS. **8**A to **8**J. The rotation of the first end **462** of the bogie **460** out of the indent **422** causes the bogie **460** to rotate about the second pivot **484**, which forces the second end **474** of the bogie lever **470** to rotate away from the hinge **440**. The rotation of the bogie lever **470** away from the hinge **440** drives rotation of the second portion **442** of the lower member **430** relative to the first portion **432** of the lower member **430** (i.e. from the second configuration shown in FIGS. **8**F to the first configuration shown in FIGS. **8**H and **8**I).

[0095] The displacement of the first end **462** of the bogie **460** out of the indent **422** means that the second portion **442** of the lower member **430** is moved from the second configuration to the first configuration prior to significant rotation of the upper member **410** relative to the first portion **432** of the lower member **430** (as shown in FIG. **8**H). Therefore, the portions **432**, **442** of the lower member **430** do not straighten until the first portion **432** has been fully rotated towards the upper member **410**, and the portions **432**, **442** are returned to the unstraightened configuration prior to significant rotation of the upper member **410** relative to the first portion **432**.

[0096] Further rotation of the lower member **430** from the angular position relative to the upper member **410** shown in FIG. **8**H to the angular position relative to the upper member **410** shown in FIG. **8**J results in the cable **428** being pulled back through the first end **412** of the upper member **410**, reducing compression on the spring, and thereby reducing resistance to angular rotation of the lower member **430** towards the upper member **410**. The second portion **442** of the lower member **430** stays in the second configuration during rotation of the lower member **430** away from the upper member **410** because the rotation of the bogie **460** is restricted by the substantially constant radius surface **424** of the cam **420** and the stop **466**.

[0097] FIGS. **9**A to **9**F show a fifth embodiment of a body weight support **500**. The body weight support **500** includes an upper member **510** having a first end **512** housing a first pulley **526**. The body weight support **500** also includes a lower member **530** comprising a first portion **532** and a second portion **542**. The first portion **532** of the lower member **530** is pivotally coupled to the first end **512** of the upper member **510** about a first pivot **580** and pivotally coupled to the second portion **542** of the lower member **530** about a hinge **540**. A second pulley **548** is coincident with the hinge **540**.

[0098] The first portion **532** of the lower member **530** also includes a stop **556** that protrudes outwardly from the first portion **532** so that it is capable of abutting the second portion **542** when the second portion **542** is rotated into contact with it (i.e. as shown in FIG. **9**A). The stop **556** limits the extent of rotation of the second portion **542** away from the upper member **510** about the hinge **540**.

[0099] As with the fourth embodiment, the body weight support **500** includes a load bearing element comprising a replaceable cable **528**. A first end **538** of the cable **528** is attached to the second portion **542** of the lower member **530** such that when the second portion **542** abuts the stop **556**, the tangential distance between the hinge **540** and the cable **528** exceeds the radius of the first pulley **526**. This means that, when the body weight support **500** is in the configuration shown in FIG. **9**A, the cable **528** is not in contact with the second pulley **548** (i.e. there is a gap between the cable **528** and the second pulley **548**).

[0100] The load bearing element also includes a spring (not shown), which is housed within the upper member **510**. A second end (not shown) of the cable **528** is attached to the end of the spring that is furthest from the first pulley **526**, meaning that the cable **528** protrudes from an opening in the first end **512** of the upper member **510**. The spring is compressed when the cable **528** is pulled out of the first end

512 (i.e. as the lower member **530** is rotated relative to the upper member **510** from the angular position shown in FIG. **9**A to the angular position shown in FIG. **9**C, for example).

[0101] The force applied by the cable **528** to the second portion **542** of the lower member **530** pulls the second portion **542** of the lower member **530** away from the upper member **510**. As mentioned above, the stop **556** limits the extent of rotation of the second portion **542** away from the upper member **510**.

[0102] When the body weight support **500** is in the configuration shown in FIG. **9**A, the cable **528** is in contact with the first pulley **526** and is not in contact with the second pulley **548**. This means that when the lower member **530** (specifically, the second portion **542** of the lower member **530**) is rotated relative to the upper member **510** to the angular position shown in FIG. **9**B, the lower member **530** pivots about the first pivot **580** and not the hinge **540**.

[0103] Further rotation of the lower member **530** towards the upper member **510** brings the first portion **532** of the lower member **530** into contact with the upper member **510** (as shown in FIG. **9**C). At this point, the first portion **532** of the lower member **530** cannot rotate further towards the upper member **510**, meaning that further rotation of the second portion **542** of the lower member **530** towards the upper member **510** causes rotation of the second portion **542** relative to the first portion **532** about the hinge **540**. This rotation moves the second portion **542** out of contact with the stop **556** and brings the cable **528** into contact with the second pulley **548** (FIG. **9**D). Further rotation of the second portion **542** relative to the first portion **532** about the hinge **540** results in movement of the second portion **542** from the first configuration relative to the first portion **532** shown in FIGS. **9**A to **9**C to the second configuration relative to the first portion **532** shown in FIG. **9**F.

[0104] In the first, second and fourth embodiments of the body weight support, a user is unable to rotate the second portion of the lower member relative to the first portion of the lower member until the lower member is in a particular angular position relative to the upper member. In contrast, a user of the body weight support **500** of the fifth embodiment is able to move the second portion **542** of the lower member **530** relative to the first portion **532** of the lower member **530** before the lower member **530** is in a particular angular position relative to the upper member **510**. This could be done, for example, by the user holding the first portion **532** and the second portion **542** and rotating them relative to one another. However, when the body weight support **500** of the fifth embodiment is in use, the rotation of the second portion **542** of the lower member **530** relative to the upper member **510** does not cause the second portion **542** of the lower member **530** to rotate relative to the first portion **532** of the lower member **530** until the lower member **530** (specifically the first portion **532** of the lower member **530**) is in a particular position (shown in FIG. **9**C) relative to the upper member **510**. The same is also true of the body weight supports of the first, second and fourth embodiments.

[0105] In the body weight supports **400**, **500** of the fourth and fifth embodiments, the load bearing element may be deactivated by decoupling the second end of the cable **428**, **528** from the spring housed within the upper member **410**, **510**. This allows the cable **428**, **528** to move freely relative to the spring, meaning that the spring does not resist relative movement of the lower member **430**, **530** and the upper member **410**, **510**. The load bearing element can be reactivated by recoupling the second end of the cable **428**, **528** to the spring.

[0106] The singular terms "a" and "an" should not be taken to mean "one and only one". Rather, they should be taken to mean "at least one" or "one or more" unless stated otherwise. The word "comprising" and its derivatives including "comprises" and "comprise" include each of the stated features, but does not exclude the inclusion of one or more further features. The terms "upper" and "lower" should not be taken to limit the positions of the features that they describe. Instead, such terms are used for the reader's convenience and by way of example only. Skilled persons will appreciate that the devices described herein may be used in a number of different orientations in which the relative positions of the features differ from the positions described herein using the terms "upper" and "lower".

[0107] The above implementations have been described by way of example only, and the described implementations are to be considered in all respects only as illustrative and not restrictive. It will be appreciated that variations of the described implementations may be made without departing from the scope of the invention. It will also be apparent that there are many variations that have not been described, but that fall within the scope of the appended claims.

1. A body weight support, comprising:
a first member configured for location against a first portion of a limb of a user, the first portion of the limb being on a first side of a joint of the user;
a second member configured for location against a second portion of the limb on a second side of the joint of the user, wherein the second member is rotatable relative to the first member about a pivot, the second member comprising:
a first portion proximate to the pivot; and
a second portion extending from the first portion, distal to the pivot;
wherein the second portion is movable from a first configuration in which a first angle between the first portion and the first member is less than a second angle between the second portion and the first member, to a second configuration in which the difference between the first angle and the second angle is less than the difference between the first angle and the second angle in the first configuration;
wherein the second member is rotatable from a first angular position relative to the first member to a second angular position relative to the first member, wherein during rotation of the second member from the first angular position to the second angular position, the second portion is prevented from moving relative to the first portion until the second member is in the second angular position; and
a load bearing element configured to provide a resistance to rotation of the second member relative to the first member.

2. The body weight support according to claim **1**, wherein rotation of the second portion of the second member relative to the first member does not cause movement of the second portion from the first configuration to the second configuration when the second member is in the first angular position, and wherein rotation of the second portion of the second member relative to the first member causes movement of the second portion from the first configuration to the second configuration when the second member is in the second angular position.

3. The body weight support according to claim **1**, wherein when the second member is in the second angular position, an angle between the first portion and the first member is smaller than when the second member is in the first angular position.

4. The body weight support according to claim **3**, wherein when the second member is in the second angular position, the first portion is substantially parallel to the first member.

5. The body weight support according to claim **1**, wherein when the second portion is in the second configuration, the second portion is in line with the first portion.

6. The body weight support according to claim **1**, wherein the second portion is hingedly coupled to the first portion.

7. The body weight support according to claim **1**, wherein the length of the second portion is adjustable.

8. The body weight support according to claim **1**, wherein the length of the first portion is adjustable.

9. The body weight support according to claim **1**, wherein an angle between the first portion and the second portion in the first configuration of the second member is adjustable.

10. The body weight support according to claim **1**, further comprising a pivotal coupling configured to permit the rotation of the second member relative to the first member.

11. (canceled)

12. The body weight support according to claim **1**, wherein the load bearing element is deactivatable to permit substantially free rotation of the second member relative to the first member.

13. The body weight support according to claim **1**, wherein the first member is configured for location against an upper leg of the user and the second member is configured for location against a lower leg of the user.

14. The body weight support according to claim **13**, wherein the first member is configured for location against the outer side of the user's upper leg.

15. The body weight support according to claim **13**, wherein the second portion is connectable to a boot of the user.

16. The body weight support according to claim **15**, wherein the second portion is connectable to the rear of the boot of the user.

17. The body weight support according to claim **1**, further comprising a connection member pivotally coupled to the first portion of the second member, wherein the connection member engages an end of the load bearing element.

18. The body weight support according to claim **17**, wherein the length of the connection member is adjustable.

19. The body weight support according to claim **17**, wherein an end of the connection member is slidably movable within a slot in the first member, and wherein the end of the connection member engages the end of the load bearing element.

20. The body weight support according to claim **1**, wherein the load bearing element comprises a cable.

21. The body weight support according to claim **20**, wherein a first end of the cable is attached to the second portion of the second member.

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